

# Calibration Behavior: 2024 vs. 2026

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## Introduction

This report compares longitudinal calibration behavior between the 2024 and 2026 calibration datasets retrieved from RIPE Atlas Anchors. The analysis reproduces the drift analysis originally applied to the 2024 calibration data by Prof. Zach Weinberg of Carnegie Mellon University. For both the 2024 and 2026 data, calendar dates are preserved throughout the analysis. Thus, the `days` variable represents actual elapsed calendar days rather than the number of calibration files retained after missing days were removed.

Two measures of drift are examined:

1. **Maximum drift:** the largest instantaneous drift observed up to each point in time, with respect to the calibration on Day 0.
2. **Instantaneous drift:** the absolute proportional difference between an anchor's current calibration and its calibration on Day 0.

## Load Data

The 2026 data requires us to calculate calibration value via given slope and intercept values.

The calibration value is calculated as:

$$y = 100m + b$$

where (m) is the slope and (b) is the intercept.

Table 1: Dataset Summaries

| year | anchors | observations | first_date | last_date  | calendar_days |
|------|---------|--------------|------------|------------|---------------|
| 2024 | 659     | 19111        | 2024-10-21 | 2024-11-18 | 28            |
| 2026 | 444     | 35520        | 2026-01-23 | 2026-04-14 | 81            |

Only anchors with complete observations across all retained calibration dates are used in the final comparison, despite the attempt to preserve all anchors with 0.8% retention across all files in the previous report *Anchor Representation: 2024 vs. 2026* available in the repository.

## Calculating Drift

The following calculation is equivalent to Professor Weinberg's original 2024 analysis.

For each anchor:

$$drift_t = \frac{|calibration_t - calibration_0|}{calibration_0}$$

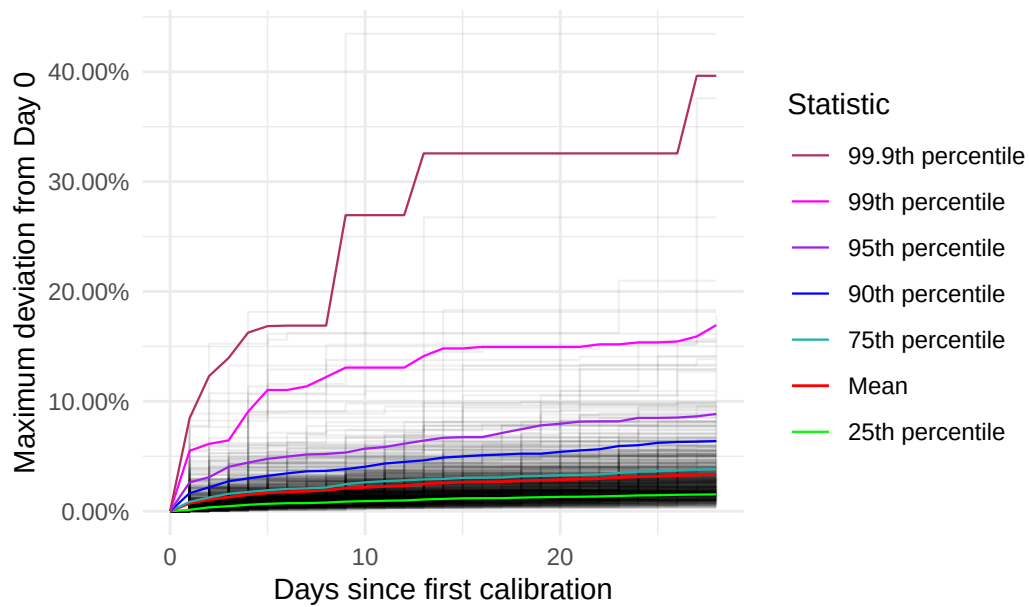
# Overview

The report is organized as follows:

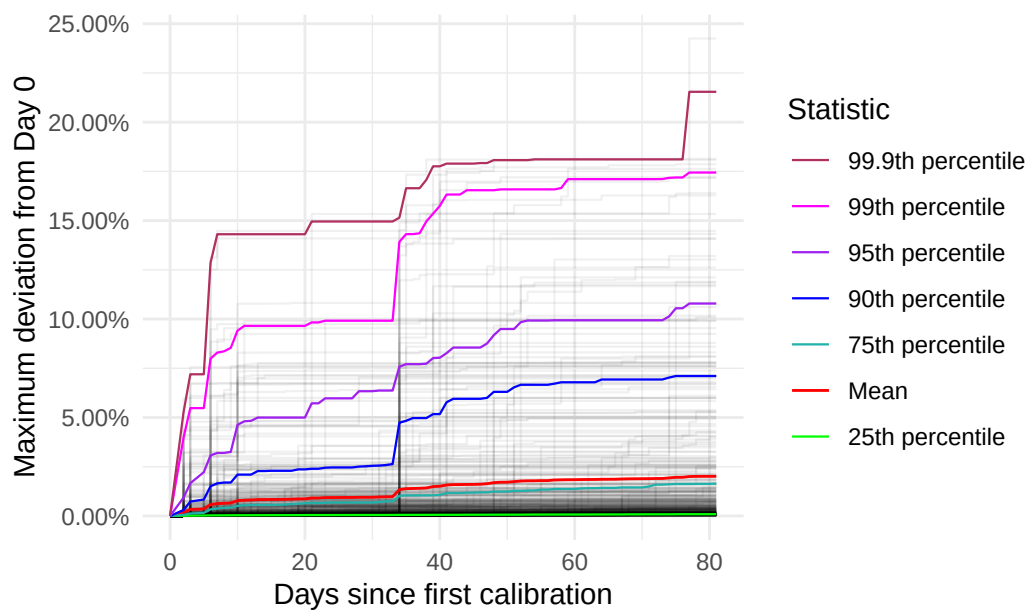
1. Maximum Drift
  1. 90-99.9th Percentile Anchor Summary
  2. Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range
  3. Visualizing ID Behavior of Repeated ASNs
  4. Flaw with Maximum Calibration Drift Visualization: Day 0 as Ground Truth
  5. Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range
  6. Filtered Maximum Drift (Day 0 Adjustment)
  7. Discussion on Adjusted Maximum Drift Visualization
2. Maximum Drift Threshold
  1. Observations/Discussion
  2. Anchor IDs that Exceeded 20%, 10%, and 5%
3. Instantaneous Drift
  1. 90-99.9th Percentile Anchor Summary
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  3. Visualizing ID Behavior of Repeated ASNs
  4. Flaw with Maximum Calibration Drift Visualization: Day 0 as Ground Truth
  5. Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range
  6. Filtered Instantaneous Drift (Day 0 Adjustment)
  7. Discussion on Adjusted Instantaneous Drift Visualization
4. Instantaneous Drift Threshold
  1. Observations/Discussion
  2. Anchor IDs that Exceeded 20%, 10%, and 5%
5. Direct 2024 vs. 2026 Comparison
  1. Maximum Calibration Drift
  2. Instantaneous Calibration Drift
  3. Discussion
6. Notes for Interpretation/Process

## 1. Maximum Drift

2024 Maximum Calibration Drift



2026 Maximum Calibration Drift



## 1.1 90-99.9th Percentile Anchor Summary

- 653 anchors were in the range at least once in 2024 data
- 444 anchors were in the range at least once in 2026 data
- 276 anchors were in the range at least once in both 2024 and 2026 data
- 185 anchors occupied the range only one time each in both 2024 and 2026 data
- 91 anchors occupied the range more than one time each between 2024 and 2026 data

## 1.2 Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range

Below, we take a look at the 91 IDs that appeared in the 90-99.9th percentile range more than one time each between the 2024 and 2026 datasets. After joining geographic context to these anchors `cleaned_persistent_90_999_both`, we find that several ASN IDs are associated more than once with Anchor IDs consistently in the 90-99.9th percentile range.

Table 2: ASN Frequency

| asn    | anchors |
|--------|---------|
| 14061  | 4       |
| 16276  | 4       |
| 3491   | 3       |
| 57695  | 3       |
| 36236  | 2       |
| 197540 | 2       |

AS14061 belongs to DigitalOcean, LLC, registered in the United States, and hosts 49.9% of its IPv4 share in the United States, 12.4% in Germany, 10.8% in the Netherlands, and 10.3% in Singapore. The rest of the percentages reported are below 10% and are reported to be in the United Kingdom, India, Canada, and Australia. The same countries are listed with different percentages for IPv6. [Link for ASN 14061](#)

AS16276 belongs to OVH SAS, registered in France. It hosts 55.5% of its IPv4 share in France, 17.5% in Canada, and less than 10% in the United States, Germany, the United Kingdom, Poland, Singapore, and Australia. It hosts less than 1% in India, Italy Ireland, Netherlands, Belgium, and Hong Kong. The same countries are listed for IPv6, with different percentages. [Link for ASN 16276](#)

AS3491 belongs to PCCW Global (HK) Ltd., registered in Hong Kong. It hosts 32% of its IPv6 share in Hong Kong, 28.9% in South Africa, 28.6% in Mozambique, and less than 3% in the United States, Singapore, Japan, and <1% in Germany, France, Taiwan, South Korea, United Kingdom, Australia, Netherlands, Mexico, Sweden, and Kenya. The same countries occupy the shares of IPv4 addresses, with different percentages. [Link for ASN 3491](#)

AS57695 belongs to Misaka Network, Inc., and is registered in the the United States. It is registered in the United States, but does not maintain any IP addresses in the country. It hosts its IPv4 equally amongst the Netherlands, France, Singapore, Brazil, Switzerland,

and Germany at 16.7%. For IPv6 addresses, the same countries are listed, but 48.3% of the share is associated with the Netherlands, 48.2% with France, 3.1% with Singapore, and the remaining countries are associated with less than 1%. [Link for ASN 57695](#)

AS36236 belongs to NetActuate, Inc, registered with the United States. 68.6% of its IP share is in the United States, and less than 5% are in 20 different countries. The IPv6 share is similar, with the United States at 65.8% South Africa, Ghana, and Kenya above 9%, and the same IPv4 listed countries for the remaining share. [Link for ASN 36236](#)

AS197540 belongs to netcup GmbH, registered in Germany. Germany dominates its IPv4 share with 89.2%, Austria represents 9.8%, and the Netherlands the remaining 1%. The same three countries constitute the IPv6 share. [Link for ASN 197540](#)

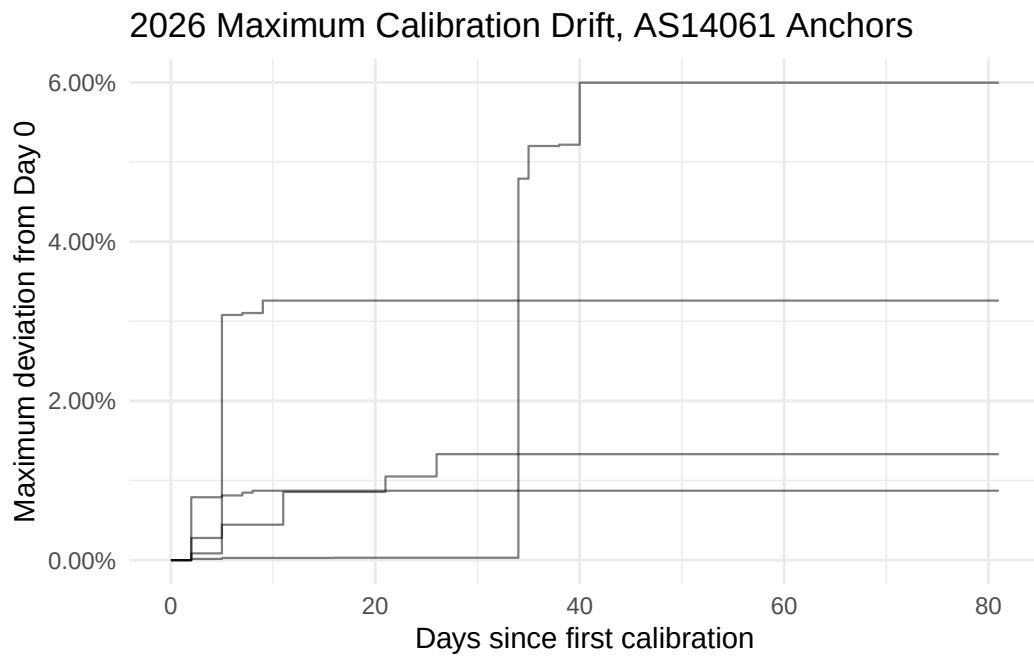
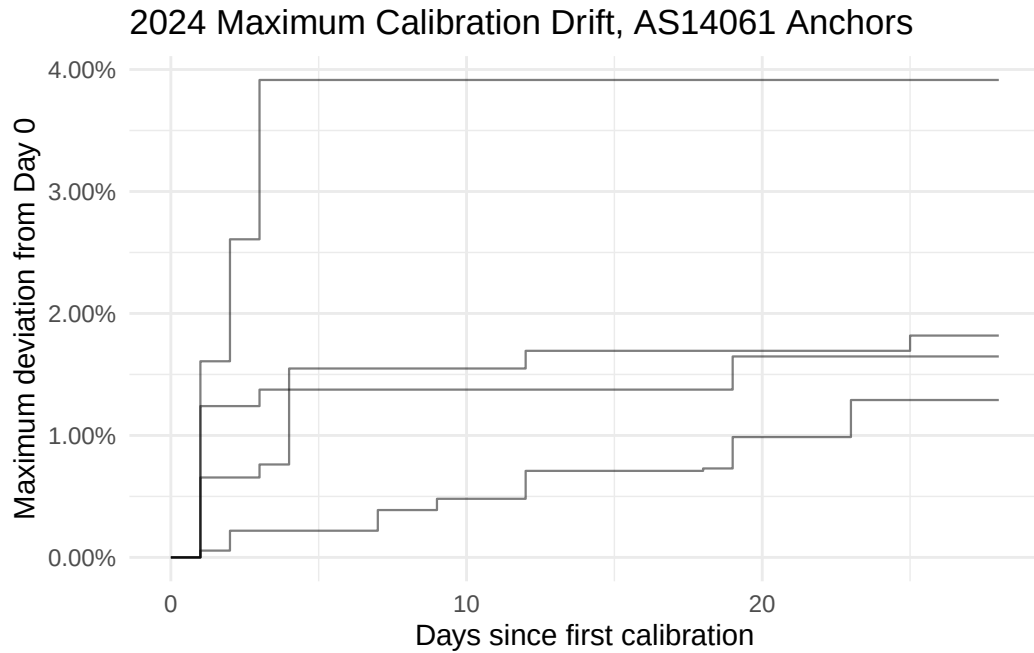
Table 3: Country Frequency

| country | anchors |
|---------|---------|
| DE      | 16      |
| US      | 14      |
| SG      | 8       |
| AT      | 7       |
| FR      | 6       |
| IT      | 6       |
| CZ      | 5       |
| GB      | 5       |
| NL      | 3       |

This table lists Germany, the United States, Singapore, Austria, France, Italy, Czechia, United Kingdom, and the Netherlands as being associated with more than 2 anchors that occupy the 90-99.9th percentile consistently. Given the IPv4 and IPv6 shares of the previously identified host network providers, this distribution of countries seems reasonable. The city and continent distributions are available in the code as well, but are omitted.

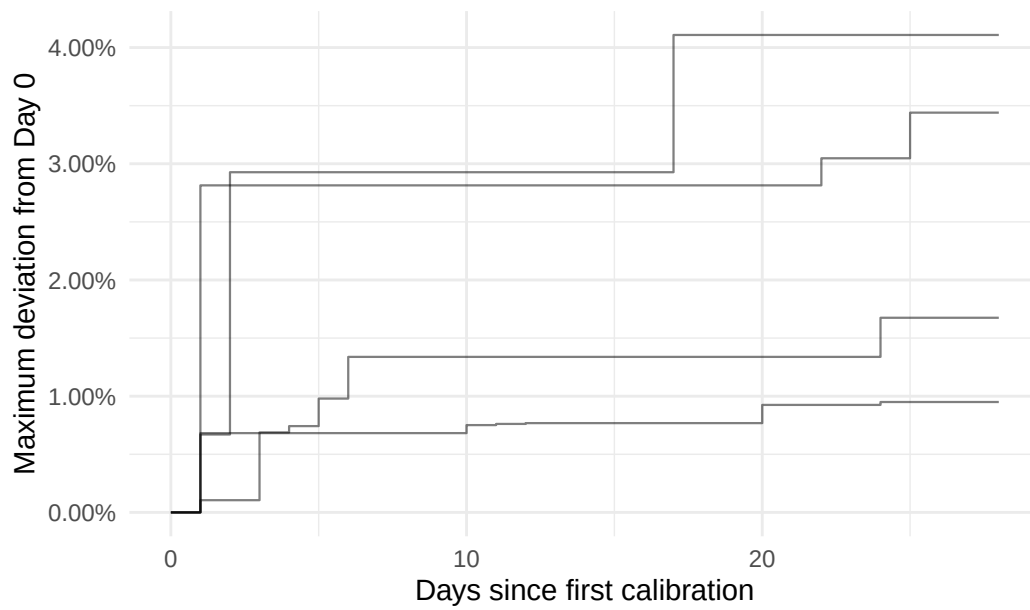
### 1.3 Visualizing Repeated ASN in 90-99.9th Percentile Range

#### 1.3.1 AS14061

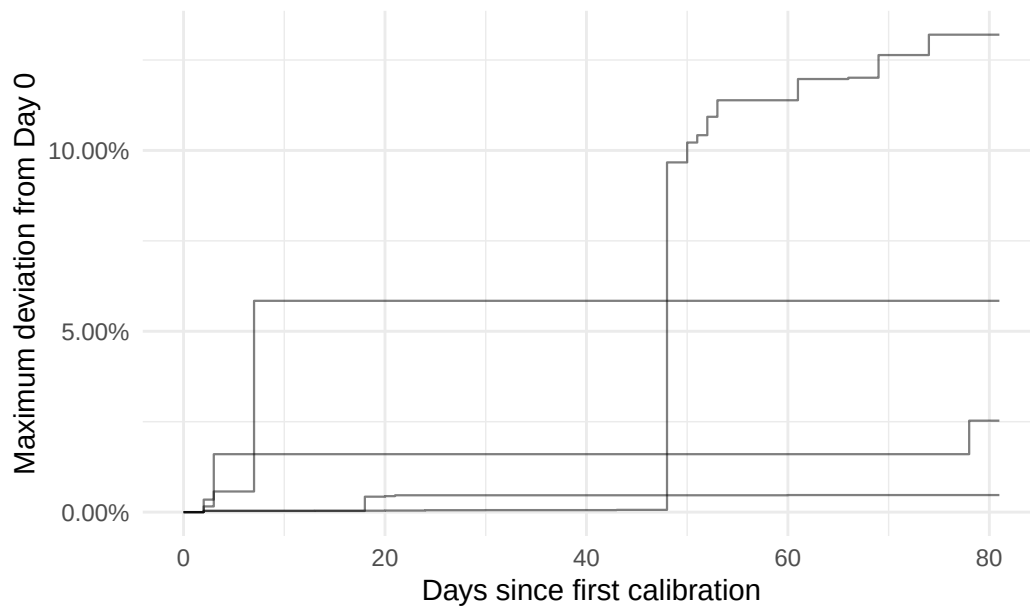


### 1.3.2 AS16276

2024 Maximum Calibration Drift, AS16276 Anchors



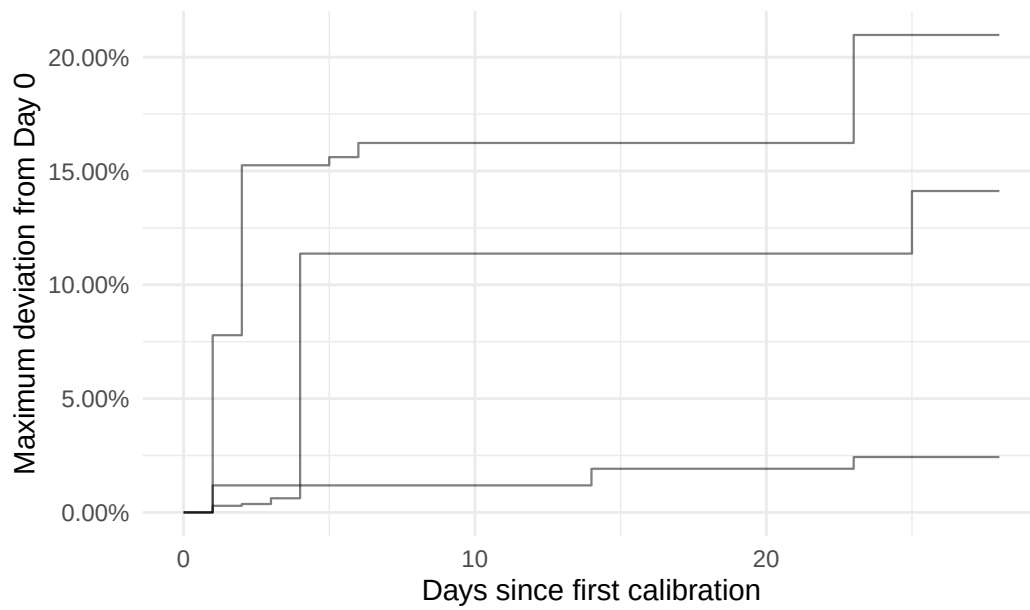
2026 Maximum Calibration Drift, AS16276 Anchors



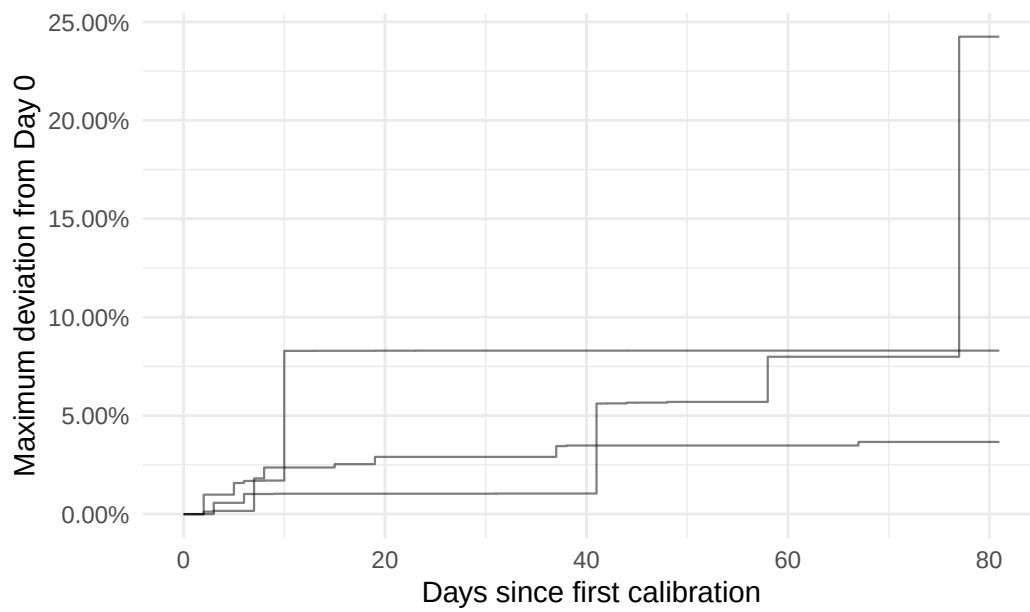


### 1.3.3 AS3491

2024 Maximum Calibration Drift, AS3491 Anchors

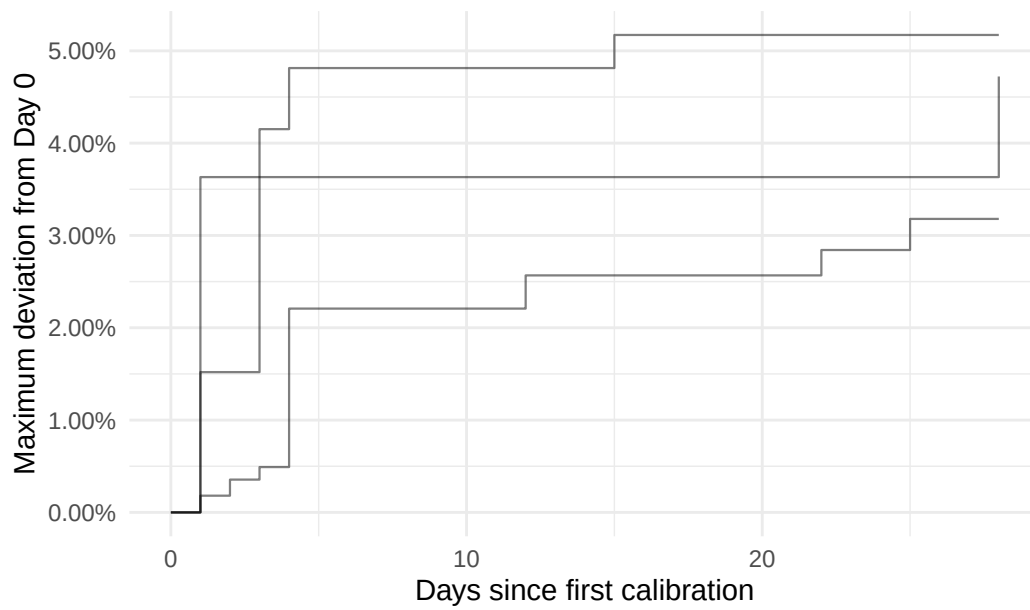


2026 Maximum Calibration Drift, AS3491 Anchors

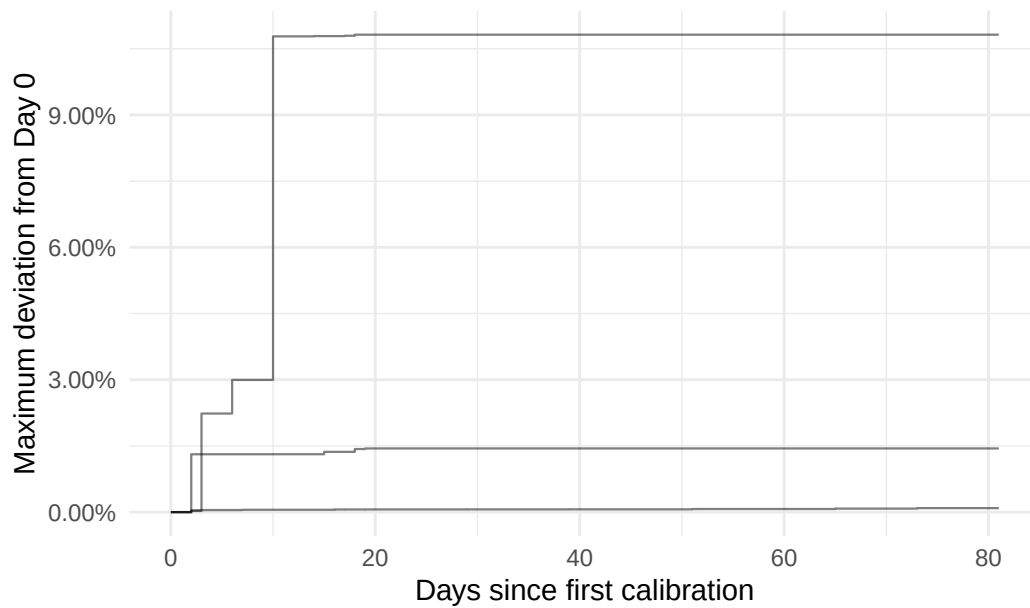


### 1.3.4 AS57695

2024 Maximum Calibration Drift, AS57695 Anchors

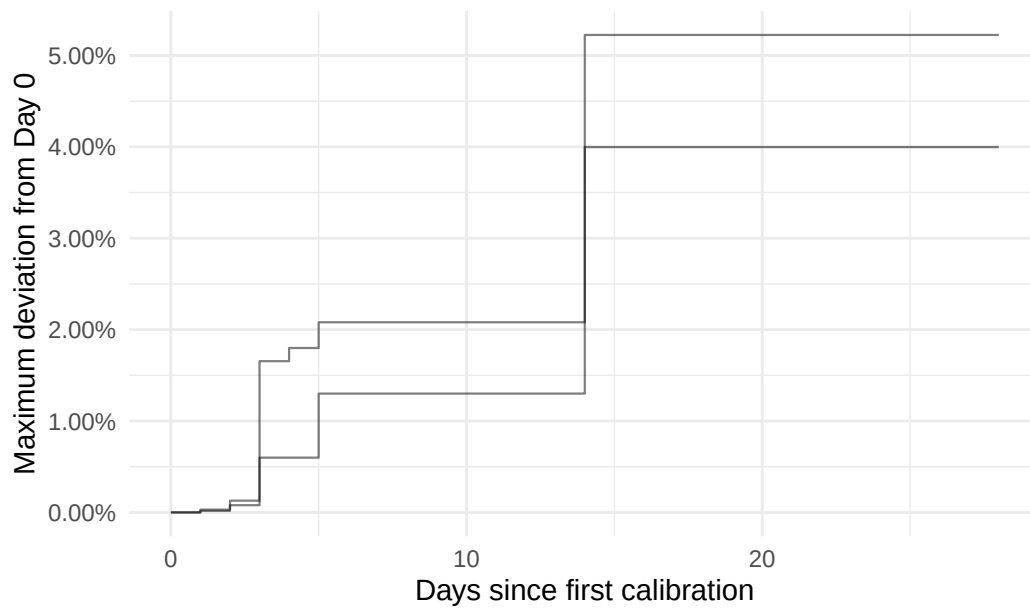


2026 Maximum Calibration Drift, AS57695 Anchors

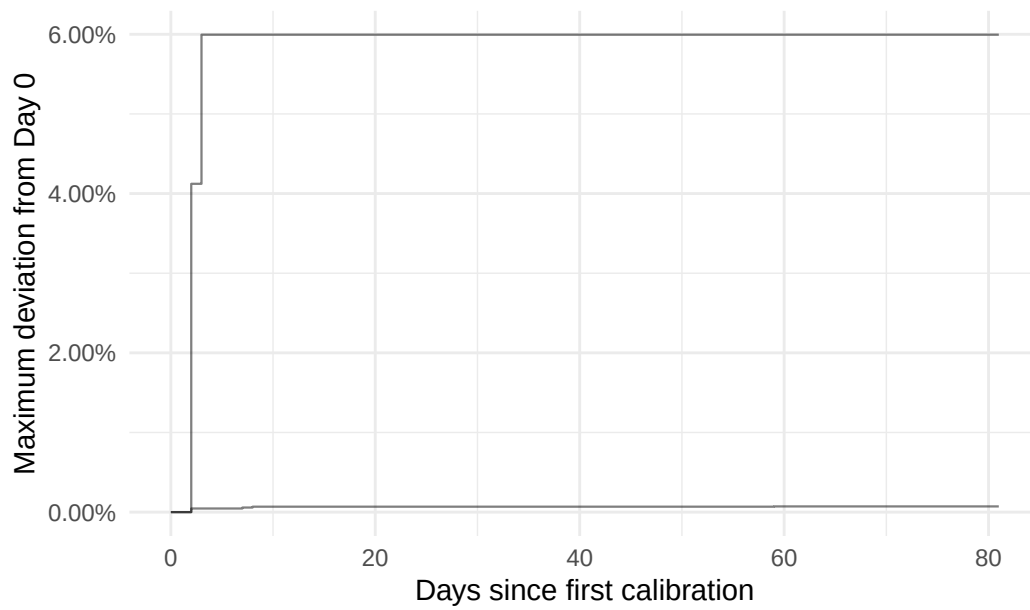


### 1.3.5 AS36236

2024 Maximum Calibration Drift, AS36236 Anchors

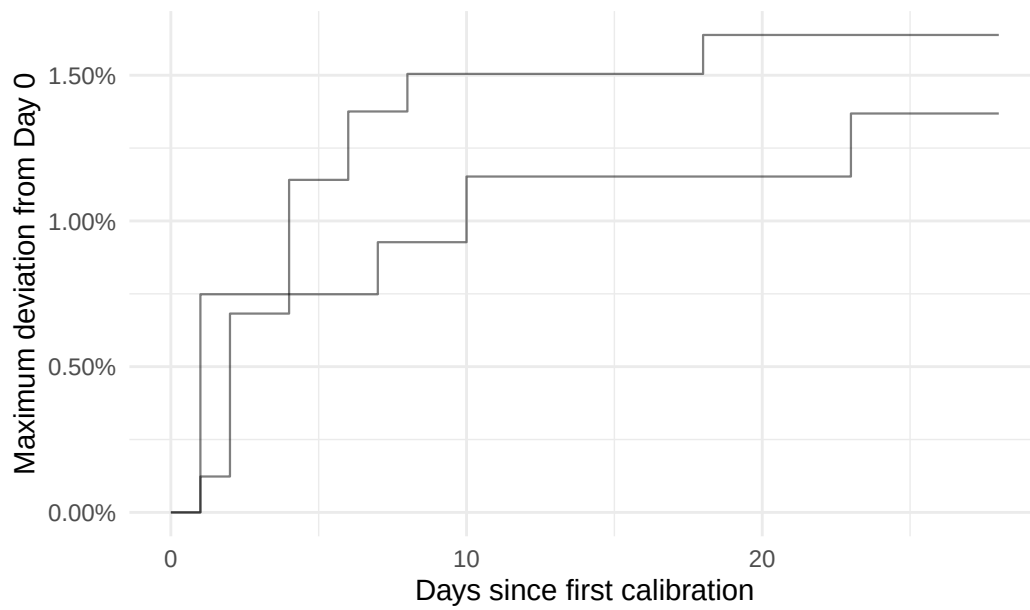


2026 Maximum Calibration Drift, AS36236 Anchors

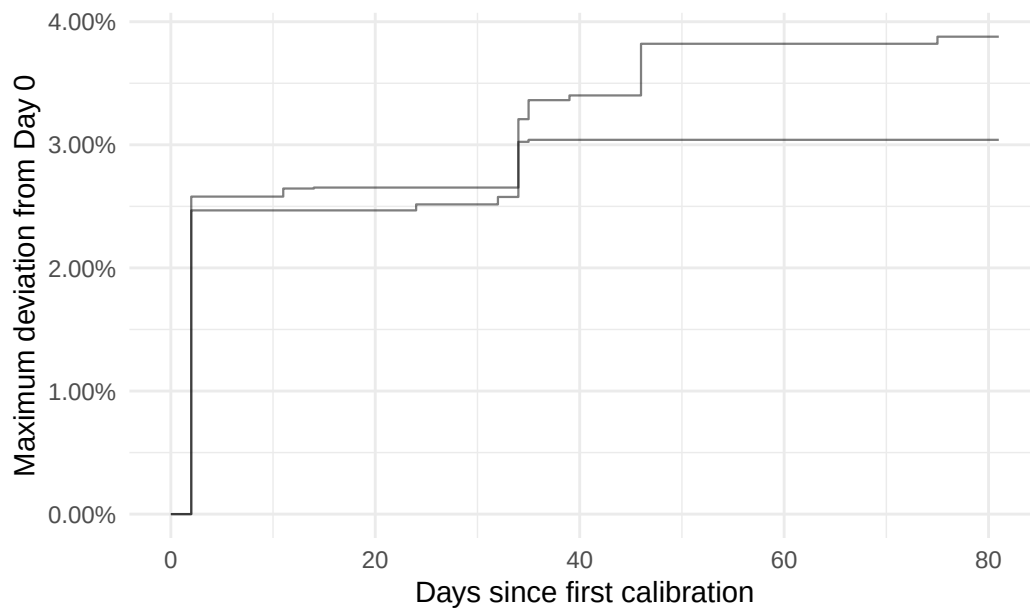


### 1.3.5 AS197540

2024 Maximum Calibration Drift, AS197540 Anchors



2026 Maximum Calibration Drift, AS197540 Anchors



## 1.4 Flaw with Maximum Calibration Drift Visualization: Day 0 as Ground Truth

Once we visualize the individual anchor behaviors associated with repeated ASNs within the 90-999 range, we find that the 5 of 6 ASNs maintain maximum drift below 12%. These 5 ASNs that appeared more than once in the 90-999 range seemed to be associated with anchor IDs with erratic behavior, but appear to be rather conservative upon visualizing their drift behavior. AS3491, however, appears to be responsible for shifting the 99.9th percentile towards 25% in 2026 and exceeds 20% drift in 2024.

One ID in AS57675 exceeds 10% drift around day 10 in 2026, but maintains this drift for the remainder of the recorded period. This is Anchor ID 7011, and is in the 90-999 range only once out of 29 days in 2024, but appears in the range 79 of 80 days in 2026. This suggests that the day 0 calibration measurement for 2026 was an outlier, and for the 79 out of 80 days, the calibration value returned to its ground truth calibration range, which was around 10% away from the calibration value recorded on day 0. The maximum drift chart is not enough information to gauge outlier IDs to further filter our optimal set of reliable IDs, as it determines drift strictly with respect to the day 0 measurement.

Take a look at the individual anchors associated with the ID Tables grouped by ASN. Highlighted are three IDs that likely entered the 90-99.9th percentile range due to the strict day 0 ground truth assumption of Maximum Drift visualization.

Table 4: AS14061 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6968 | 9           | 29       | 1           | 80      | Asia      | IN      | Bangalore |
| 6967 | 1           | 29       | 15          | 80      | Europe    | DE      | Frankfurt |
| 6970 | 1           | 29       | 30          | 80      | Asia      | SG      | Singapore |
| 6971 | 1           | 29       | 3           | 80      | Europe    | GB      | London    |

Table 5: AS16276 Anchors

| id   | in_range_24 | to-tal_24 | in_range_26 | total26 | continent | country | city              |
|------|-------------|-----------|-------------|---------|-----------|---------|-------------------|
| 6727 | 4           | 29        | 37          | 80      | Oceania   | AU      | Sydney            |
| 6800 | 3           | 29        | 1           | 80      | Europe    | PL      | Ozarów Mazowiecki |
| 6696 | 1           | 29        | 35          | 80      | Europe    | FR      | Gravelines        |
| 6726 | 1           | 29        | 4           | 80      | Asia      | SG      | Singapore         |

Table 6: AS3491 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city   |
|------|-------------|----------|-------------|---------|-----------|---------|--------|
| 7198 | 27          | 29       | 20          | 80      | Asia      | AE      | Dubai  |
| 7078 | 26          | 29       | 76          | 80      | Oceania   | AU      | Sydney |

|      |   |    |    |    |      |    |           |
|------|---|----|----|----|------|----|-----------|
| 7010 | 1 | 29 | 32 | 80 | Asia | SG | Singapore |
|------|---|----|----|----|------|----|-----------|

Table 7: AS57695 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city     |
|------|-------------|----------|-------------|---------|---------------|---------|----------|
| 6927 | 15          | 29       | 4           | 80      | North America | US      | Secaucus |
| 6928 | 8           | 29       | 1           | 80      | North America | US      | Reston   |
| 7011 | 1           | 29       | 79          | 80      | North America | US      | Seattle  |

Table 8: AS36236 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city    |
|------|-------------|----------|-------------|---------|---------------|---------|---------|
| 6379 | 5           | 29       | 1           | 80      | North America | US      | Durham  |
| 6408 | 1           | 29       | 45          | 80      | North America | US      | Phoenix |

Table 9: AS197540 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6581 | 1           | 29       | 27          | 80      | Europe    | DE      | Nuremberg |
| 6799 | 1           | 29       | 32          | 80      | Europe    | DE      | Nürnberg  |

### 1.5 Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range

Due to the generally conservative nature of the IDs associated with ASNs frequently appearing in the 90-999 range, we must not overlook the ASNs making individual appearances. Of course, we maintain AS3941 as being associated with frequent outlier drift behavior amongst the repeated ASNs. We will now examine the IDs that are associated with unique ASNs that entered the 90-999 range.

To pare down the IDs for further analysis, we will take away the likely misplaced IDs due to the day 0 ground truth procedure. These IDs will be highlighted in yellow.

Table 10: Anchors Associated with Unique ASNs

| asn    | id   | in_range_24 | total_24 | in_range_26 | total26 | country | city              |
|--------|------|-------------|----------|-------------|---------|---------|-------------------|
| 719    | 6251 | 1           | 29       | 2           | 80      | FI      | Tampere           |
| 203329 | 6313 | 1           | 29       | 76          | 80      | DE      | Frankfurt am Main |
| 48943  | 6325 | 1           | 29       | 49          | 80      | AT      | Vienna            |
| 2108   | 6327 | 1           | 29       | 25          | 80      | HR      | Zagreb            |

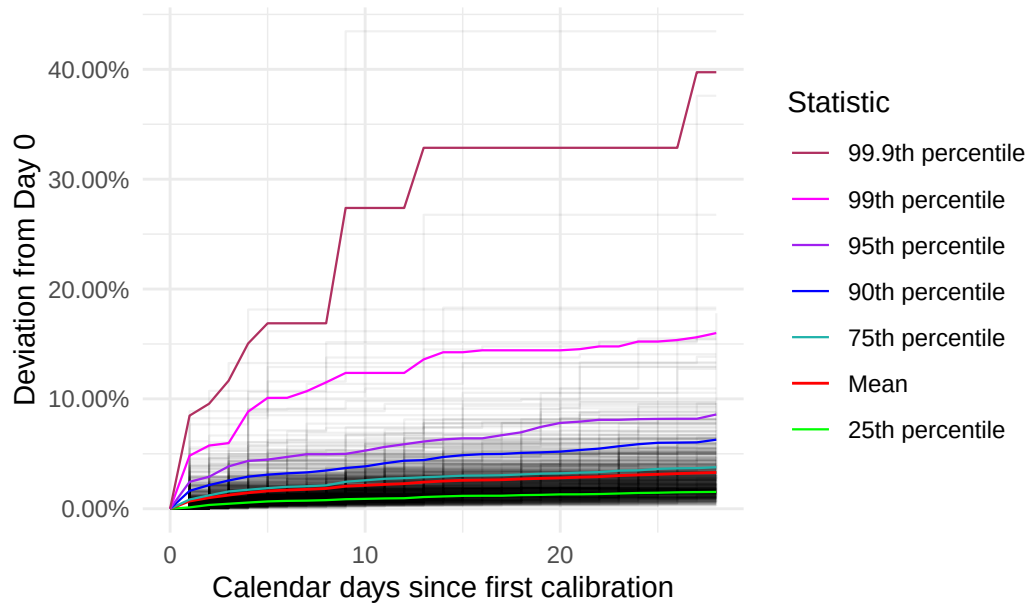
|        |      |   |    |    |    |    |                   |
|--------|------|---|----|----|----|----|-------------------|
| 62416  | 6328 | 1 | 29 | 73 | 80 | PT | Lisbon            |
| 24940  | 6332 | 1 | 29 | 14 | 80 | DE | Nuremberg         |
| 2857   | 6336 | 1 | 29 | 53 | 80 | DE | Mainz             |
| 39878  | 6360 | 1 | 29 | 49 | 80 | AT | Salzburg          |
| 41000  | 6423 | 1 | 29 | 2  | 80 | GB | Manchester        |
| 35574  | 6435 | 1 | 29 | 49 | 80 | IT | Rozzano           |
| 20473  | 6506 | 1 | 29 | 4  | 80 | SG | Singapore         |
| 31034  | 6509 | 1 | 29 | 49 | 80 | IT | Arezzo            |
| 39642  | 6545 | 1 | 29 | 3  | 80 | DK | Kolding           |
| 41095  | 6568 | 1 | 29 | 4  | 80 | SG | Singapore         |
| 39617  | 6674 | 1 | 29 | 3  | 80 | GB | Balderton         |
| 24482  | 6676 | 1 | 29 | 3  | 80 | SG | Singapore         |
| 62310  | 6751 | 1 | 29 | 2  | 80 | DE | Berlin            |
| 41327  | 6762 | 1 | 29 | 70 | 80 | IT | Palermo           |
| 46997  | 6794 | 1 | 29 | 71 | 80 | US | Seattle           |
| 24971  | 6809 | 1 | 29 | 26 | 80 | CZ | Brno              |
| 199399 | 6841 | 1 | 29 | 44 | 80 | GR | Athens            |
| 21320  | 6874 | 1 | 29 | 70 | 80 | FR | Paris             |
| 62513  | 6888 | 1 | 29 | 4  | 80 | CA | Toronto           |
| 38064  | 6904 | 1 | 29 | 5  | 80 | NZ | Auckland          |
| 39835  | 6922 | 1 | 29 | 48 | 80 | DE | Göttingen         |
| 39636  | 7003 | 1 | 29 | 36 | 80 | NL | Arnhem            |
| 30971  | 7008 | 1 | 29 | 42 | 80 | AT | Vienna            |
| 13101  | 7012 | 1 | 29 | 8  | 80 | DE | Kiel              |
| 38001  | 7091 | 1 | 29 | 2  | 80 | SG | Singapore         |
| 56381  | 7097 | 1 | 29 | 79 | 80 | DE | Frankfurt am Main |
| 21034  | 7110 | 1 | 29 | 2  | 80 | IT | Pescara           |
| 33920  | 7125 | 1 | 29 | 5  | 80 | GB | Leeds             |
| 29423  | 7141 | 1 | 29 | 31 | 80 | DE | Frankfurt         |
| 50030  | 7156 | 1 | 29 | 28 | 80 | DE | Berlin            |
| 39063  | 7169 | 1 | 29 | 49 | 80 | DE | Frankfurt         |
| 59678  | 7193 | 1 | 29 | 33 | 80 | US | Piscataway        |
| 60841  | 7201 | 1 | 29 | 4  | 80 | NL | Amsterdam         |
| 25248  | 7211 | 1 | 29 | 16 | 80 | CZ | Prague            |
| 35492  | 7216 | 1 | 29 | 49 | 80 | AT | Vienna            |
| 54316  | 7224 | 1 | 29 | 2  | 80 | US | Greenwich         |
| 29287  | 7225 | 1 | 29 | 49 | 80 | AT | Vienna            |
| 23428  | 7238 | 1 | 29 | 61 | 80 | US | Kansas City       |
| 14907  | 7261 | 1 | 29 | 60 | 80 | NL | Amsterdam         |
| 200187 | 7265 | 1 | 29 | 49 | 80 | DE | Frankfurt am Main |
| 216444 | 7272 | 1 | 29 | 28 | 80 | SG | Singapore         |
| 32167  | 7292 | 1 | 29 | 2  | 80 | HK | Hong Kong         |
| 36444  | 6389 | 2 | 29 | 1  | 80 | US | Detroit           |
| 48362  | 6476 | 2 | 29 | 1  | 80 | AT | Feldkirch         |

|        |      |    |    |    |    |    |                          |
|--------|------|----|----|----|----|----|--------------------------|
| 204092 | 6494 | 2  | 29 | 1  | 80 | FR | Rennes                   |
| 202297 | 6547 | 2  | 29 | 1  | 80 | CZ | Prague                   |
| 199188 | 6559 | 2  | 29 | 1  | 80 | GB | London                   |
| 395823 | 7043 | 2  | 29 | 78 | 80 | US | Seattle                  |
| 31510  | 7083 | 2  | 29 | 1  | 80 | AT | Innsbruck                |
| 8473   | 7127 | 2  | 29 | 1  | 80 | SE | Malmö                    |
| 24840  | 7196 | 2  | 29 | 49 | 80 | DE | Frankfurt                |
| 60396  | 6441 | 3  | 29 | 1  | 80 | SE | Lund                     |
| 59715  | 6496 | 3  | 29 | 1  | 80 | IT | San Benedetto del Tronto |
| 199610 | 6649 | 3  | 29 | 32 | 80 | US | Los Angeles              |
| 29169  | 6363 | 4  | 29 | 1  | 80 | FR | Saint-Denis              |
| 201333 | 6675 | 5  | 29 | 1  | 80 | IT | Piacenza                 |
| 15699  | 7086 | 6  | 29 | 2  | 80 | ES | Alcalá de Henares        |
| 47295  | 7096 | 6  | 29 | 1  | 80 | DE | Bautzen                  |
| 25192  | 7223 | 6  | 29 | 1  | 80 | CZ | Prague                   |
| 7752   | 7221 | 8  | 29 | 79 | 80 | US | Seattle                  |
| 6881   | 7024 | 9  | 29 | 1  | 80 | CZ | Prague                   |
| 209097 | 6736 | 12 | 29 | 1  | 80 | FR | Trevoux                  |
| 53698  | 7159 | 14 | 29 | 48 | 80 | US | Phoenix                  |
| 13303  | 7213 | 15 | 29 | 1  | 80 | RS | Nis                      |
| 12008  | 6954 | 19 | 29 | 36 | 80 | IN | Mumbai                   |
| 16347  | 6443 | 22 | 29 | 1  | 80 | FR | Maxeville                |
| 15802  | 6962 | 26 | 29 | 1  | 80 | AE | Dubai                    |
| 202196 | 6639 | 28 | 29 | 1  | 80 | HK | Hong Kong                |
| 206186 | 6323 | 29 | 29 | 1  | 80 | PL | Katowice                 |

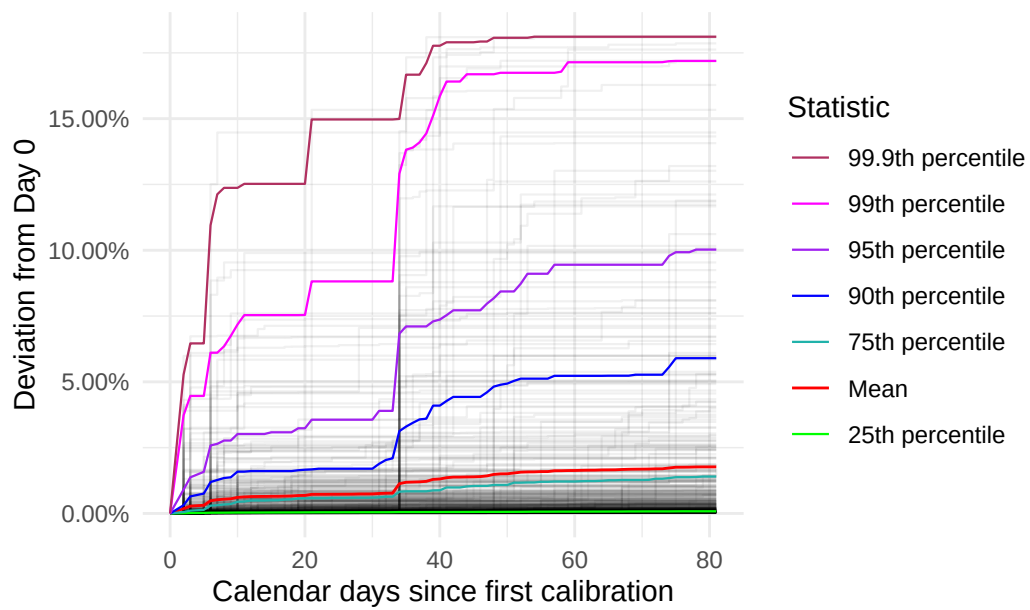


## 1.6 Filtered Maximum Drift (After Day 0 Adjustment)

### 2024 Maximum Calibration Drift



### 2026 Maximum Calibration Drift



## 1.7 Discussion on Adjusted Maximum Drift Visualizations

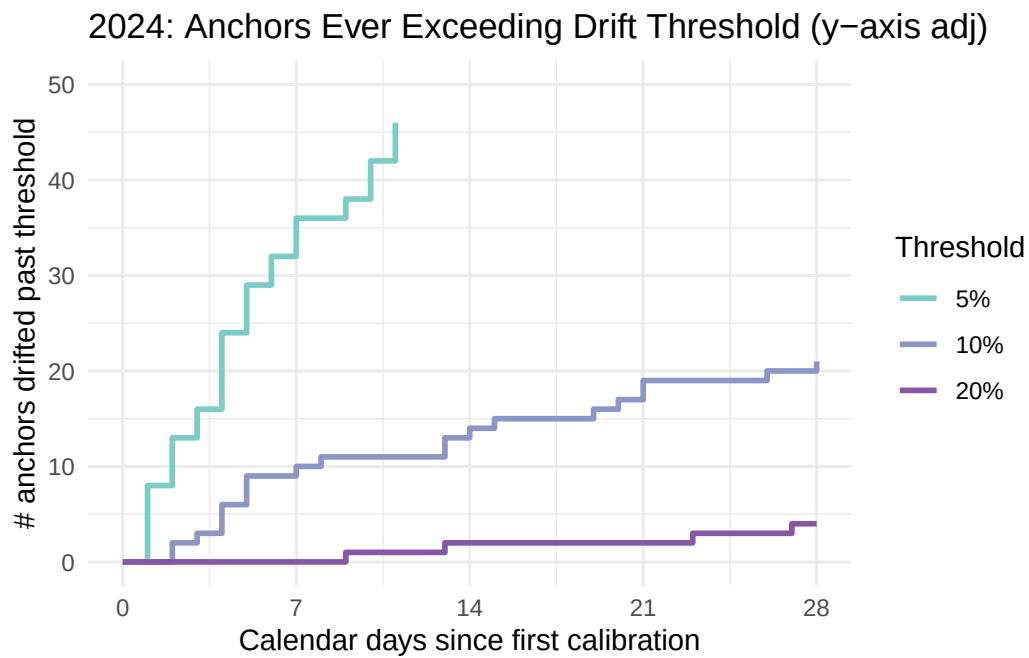
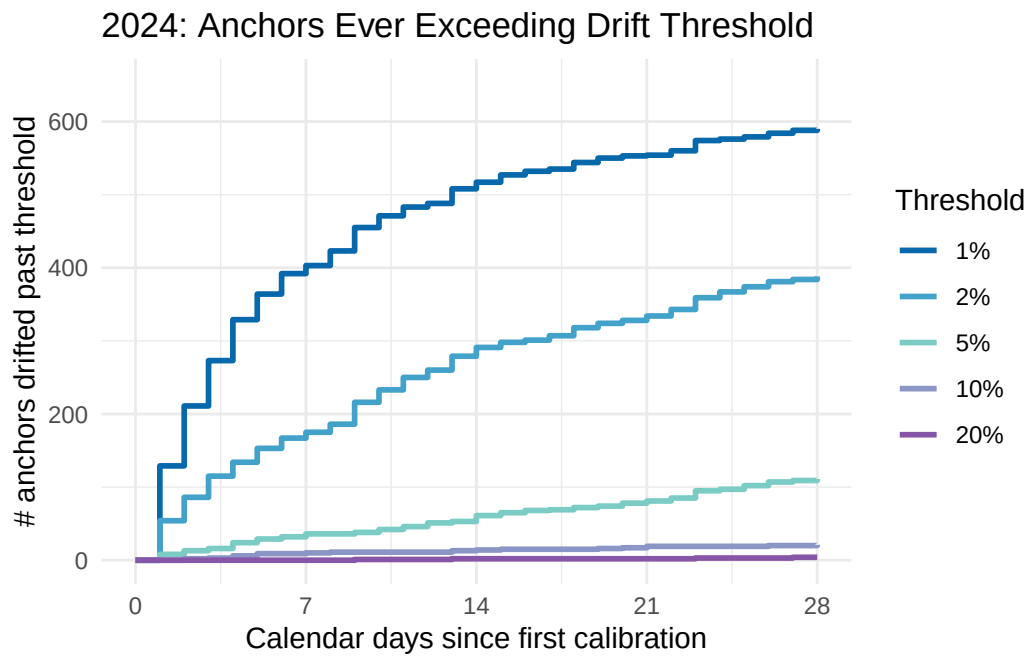
Upon identifying which anchors appeared the most in the 90-99.9th percentile range, we found that 17 of 91 anchors spent nearly all days of the recorded period in the range for one year, while only appearing the range once the other year. Furthermore, Anchor 7078 recorded 90-99.9th percentile drift 26 of 29 days in 2024 and 76 of 80 days in 2026. As the maximum calibration drift plot evaluates percentage of drift strictly with respect to the day 0 measurement, if anchors are to be at an outlier calibration value on the first day of the recorded period, a prominent drift would likely be maintained for the rest of the period as the calibration returns to its usual value. This suggests that the initial calibration values of 18 anchors including Anchor 7078 were misrepresented on day 0, and brought back to their average rates when appearing in the 90-99.9th percentile range.

We control for this flaw by removing the anchors that appear in the range more than 75% of the time in one year, while not appearing very few times in the other, or if the anchor appears in range for more than 75% of the days for both years. Ideally, we are then left with a more accurate representation of the erratic anchors whose calibration values drift significantly from day to day.

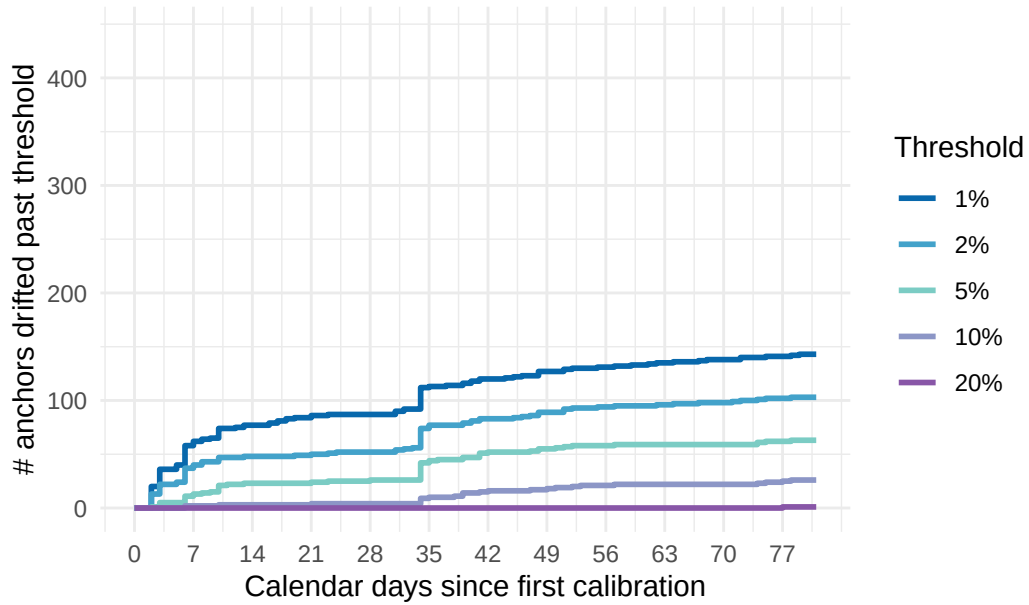
After removing the aforementioned 17 anchors, we plot maximum drift once again in 3.5.1. As the majority of the anchors removed recorded their majority 90-99.9th percentile range days in 2026 and spent 1 to at most 8 days in range in 2024, the adjustment from the original 2026 drift from 0-25% to 0-18% is indeed more prominent than the 2024 plot that maintained its 0-45% drift. Indeed, this is a very granular process, and we benefit from considering other visualizations to better understand anchor behavior.

Using Zach's 2024 analysis as a reference once again, the Maximum Drift Threshold Visualization offers another perspective in understanding our data. The following visualizations retain all anchor IDs and ASNs that were removed in this section, and represents the same data as the Maximum Drift Visualizations in Section 1. Maximum Drift.

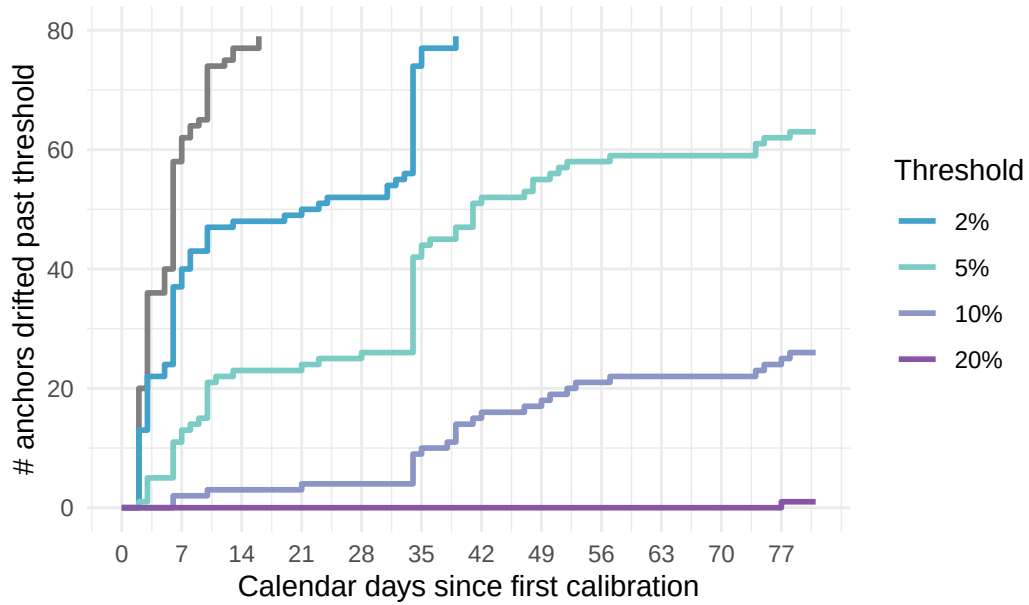
## 2. Maximum Drift Threshold



2026: Anchors Ever Exceeding Drift Threshold



2026: Anchors Ever Exceeding Drift Threshold (y-axis adj)



## 2.1 Observations/Discussion

The Maximum Drift Threshold uses percentage thresholds to illustrate the rate at which the anchors exceed 1%, 2% 5%, 10%, and 20% drift. By day 7, nearly 400 of 659 anchors in the 2024 data have exceeded 1% drift, and around 200 exceed 2% and 55 exceed 5% by day 9 and 14 respectively. In order to improve interpretability, I have adjusted the y axis for examining the 10% and 20% specific thresholds trends. In this figure, we see that about 20 of 659 anchors exceeded 10% drift, and 4 of 659 exceeded 20% by the end of the recorded period. This visualization is especially convenient for observing the rate at which the anchors collectively exceed each threshold.

For the 2026 data, we use two figures to visualize our threshold details carefully once more. The drift trends for the 2026 data appear to be more conservative than the 2024 data, which can be credited to the comparatively wider date range and therefore less consistent calibration retrieval (leaving us with 444 of 899 anchors). Nonetheless, this shift that we have been noticing around day 35 persists, this time quantifiable as around 30 new anchors reaching 2%, and 20 additional anchors reaching 5% drift thresholds. It appears that a total of around 25 anchors reach the 10% threshold by day 77, and only 1 anchor exceeds 20% over the entire recorded period.

It is encouraging to observe that in the 2024 data, the number exceeding the 20% threshold does not exceed 4 anchors, and in 2026, it is merely 1. Our previous plot appeared failed to visually quantify the number of anchors accumulating the most drift, and to understand the ratio of anchors drifting past a 20% threshold as 4 of 659 anchors is encouraging in grasping the proportion of “erratic” or “volatile” anchors to “stable” anchors that maintain 1-5%.

## 2.2 Anchor IDs that Exceeded 20%, 10%, and 5%

In 2024: 4 Anchor IDs Exceeded 20%: 7192, 7198, 7207, 7222

In 2024: 21 Anchor IDs Exceeded 10%: 6392, 6398, 6410, 6586, 6718, 6782, 6882, 6887, 6935, 6977, 6994, 7050, 7077, 7078, **7192**, **7198**, **7207**, 7219, **7222**, 7239, 7251

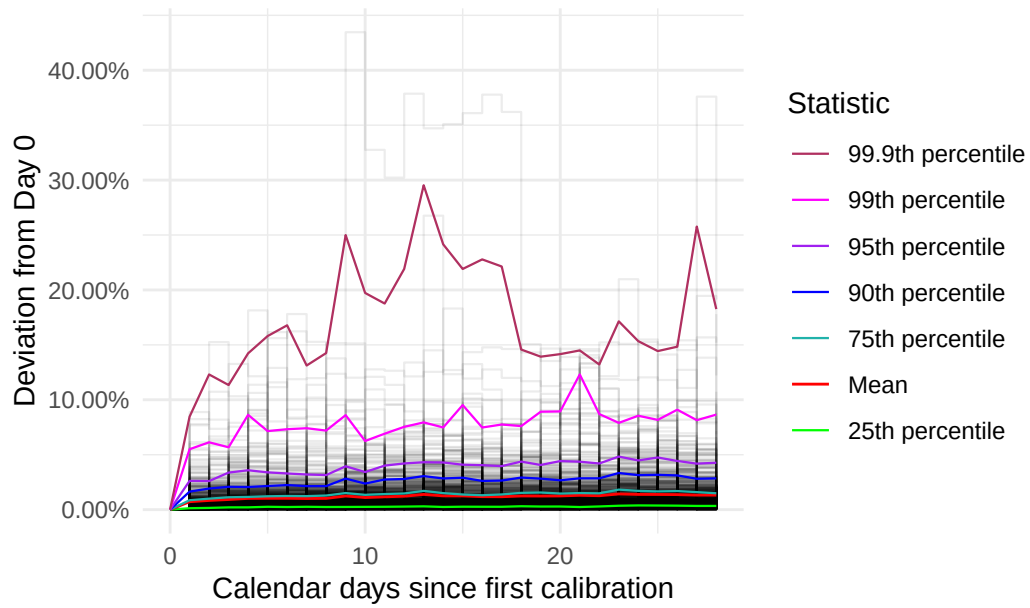
In 2026: 1 Anchor ID Exceeded 20%: 7198

In 2026: 26 Anchor IDs Exceeded 10%: 6313, 6325, 6360, 6509, 6696, 6809, 6841, 6922, 6954, 7003, 7008, 7011, 7097, 7128, 7169, 7181, **7198**, 7216, 7225, 7456, 7502, 7511, 7574, 7616, 7635, 7680

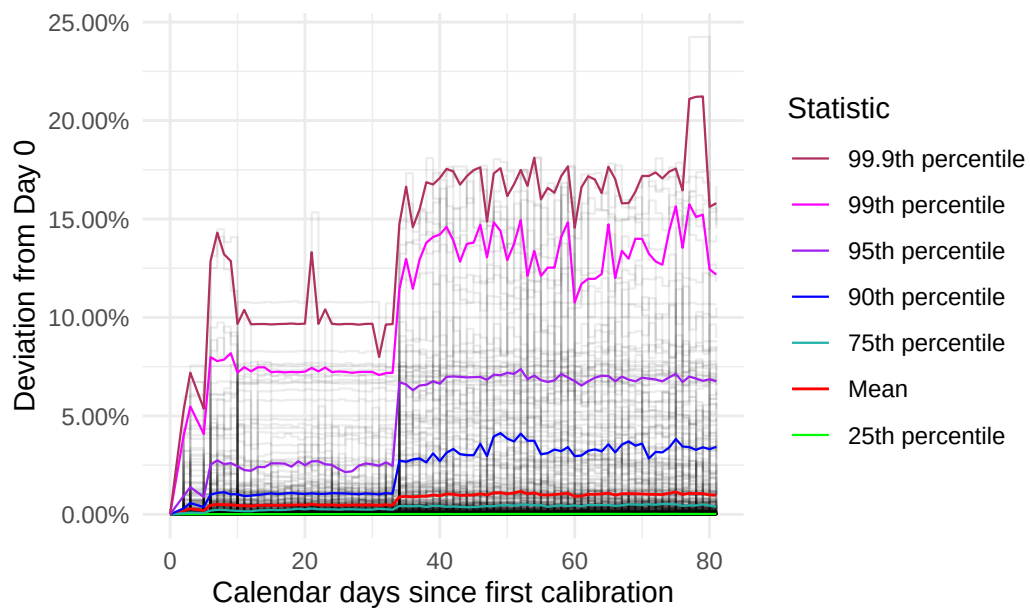
We previously discussed anchor ID 7198 in Table 6, concerning anchors associated with AS3491. The anchor occupied the 90-99.9th percentile range for 27 of 29 days in 2024, and 20 of 80 days in 2026. The other IDs 7192, 7207, and 7222 that exceeded 20% in 2024 and do not appear in the ID summaries from *Section 3.4* and *Section 3.5*. The italicized anchors are all omitted from the *Section 3.4* and *Section 3.5* summaries, and provide new data for us to consider in terms of a new subset of outlier anchors.

### 3. Instantaneous Drift Visualization

2024 Instantaneous Calibration Drift



2026 Instantaneous Calibration Drift



Following Zach's lead, I applied the Instantaneous Calibration Drift visualization on the 2024 and 2026 data. In the 2024 figure, anchor calibration drift is concentrated towards 0% and mainly seems to decrease in density between 2.5% to 7.5% deviation. The remaining fraying seems to happen between 7.5% and 20% deviation primarily, and a few outlier anchors extend above 20%, 30%, and 40%.

As for the 2026 data, anchor calibration drift seems to be mainly concentrated between 0% and 1%, and spreads similarly to the 2024 2.5% to 7.5% region in the 1% to 2.5% area between day 0 and day 35. This range seems to extend well into 7.5% after day 35. While the outliers drifted between 5% and 15% deviation before day 35, the outlier group is joined by more anchors and extends past 17.5% consistently between day 35 and the end of the recorded period. While the y-axis deviation scale is inconsistent between the two figures, it is clear that the anchor deviation behavior noticeably shifts in the 2026 data.

Due to the significantly lower number of anchors in the 2026 data and the differing time periods, interpreting these figures alone is not enough. If there were 215 or so more anchors in the 2026 data to correspond with the 659 anchors in the 2024 data, perhaps this shift behavior would have been camouflaged by a complement shift down around day 35. It is not clear that the 2024 figure is completely absent of these kinds of shifts either.

### 3.1 90-99.9th Percentile Anchor Summary

- 653 anchors were in the range at least once in 2024 data
- 444 anchors were in the range at least once in 2026 data
- 276 anchors were in the range at least once in both 2024 and 2026 data
- 144 anchors occupied the range only one time each in both 2024 and 2026 data
- 132 anchors occupied the range more than one time each between 2024 and 2026 data

### 3.2 Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range

Below, we take a look at the 132 IDs that appeared in the 90-99.9th percentile range more than one time each between the 2024 and 2026 datasets. After joining geographic context to these anchors `cleaned_persistent_90_999_instant_both`, we once again find that several ASN IDs are associated more than once with Anchor IDs consistently in the 90-99.9th percentile range. The ASNs highlighted in pink are ASNs that previously appeared as frequent 90-99.9th percentile range ASNs in section 1.2, concerning the Maximum Drift visualization. Below, we contextualize the remaining 8 ASNs that are not highlighted.

Table 11: ASN Frequency

| asn    | anchors |
|--------|---------|
| 3491   | 4       |
| 14061  | 4       |
| 16276  | 4       |
| 36236  | 3       |
| 57695  | 3       |
| 199524 | 3       |
| 8473   | 2       |
| 16347  | 2       |
| 20473  | 2       |
| 24940  | 2       |
| 35574  | 2       |
| 39923  | 2       |
| 197540 | 2       |
| 199610 | 2       |

AS199524 belongs to G-Core Labs S.A., and is registered in Luxembourg. The network hosts 22.9% of its IPv4 share in the United States, 15.4% in Luxembourg, 7.7% in the Netherlands, 5.2% in Singapore, and less than 5% in Germany, Japan, Brazil, South Korea, France, Kazakhstan, and 45 more countries. The IPv6 share spans the same countries, with the top proportion belonging to the United States at 15.5%, and the United Kingdom at 10.0%. [Link for AS199524](#)



AS8473 belongs to Bahnhof AB, a network registered in Sweden. 99.1% of its IPv4 share is in Sweden, with decimal percentages belonging to Norway, Finland, Denmark, and Germany. The IPv6 share holds 89.4% of shares in Sweden. [Link for AS8473](#)

AS16347 belongs to ADISTA SAS, a network based in France. 100% of its IPv4 and IPv6 share is in France. [Link for AS16347](#)

AS20473 is associated with The Constant Company, LLC, based in the United States. The network hosts 50.5% of IPv4 addresses in the United States, and hosts less than 10% in Japan, Singapore, the United Kingdom, Netherlands, and 18 other countries. The IPv6 share appears to host 52.6% in the United States, 10.9% in the United Kingdom, and less than 10% in Indonesia, Belgium, Turkey, and 19 other countries. [Link for AS20473](#)

AS24940 is associated with Hetzner Online GmbH, a network registered in Germany. The network's IPv4 share is split with 79.2% in Germany, 20.6% in Finland, and 0.1% in the Netherlands. The IPv6 split is similar, with 91.4% in Germany and 8.6% in Finland. [Link for AS24940](#)

AS35574 belongs to Brightstar Lottery S.p.A, an Italian network. Its IPv4 and IPv6 shares are located 100% in Italy. [Link for AS35574](#)

AS39923 is Unix-Solutions BV, a network based in Belgium. Much like the previous network, 100% of its IPv4 and IPv6 addresses are located in Belgium. [Link for AS39923](#)

AS199610 is associated with marbis GmbH, and is a network registered in Germany. 47.8% of its IPv4 share is located in Germany, while 34.1% is in the United States, and less than 10% each in the United Kingdom, Australia, Singapore, Japan, and Brazil. IPv6 addresses of the network exist primarily in Germany at 97.3%, and the remainder is divided among the United States, Singapore, the United Kingdom, and Australia. [Link for AS199610](#)

Table 12: Country Frequency

| country | anchors |
|---------|---------|
| US      | 22      |
| DE      | 20      |
| IT      | 10      |
| SG      | 10      |
| GB      | 9       |
| AT      | 8       |
| FR      | 8       |
| CZ      | 7       |
| ES      | 5       |
| PL      | 4       |
| SE      | 4       |
| NL      | 3       |

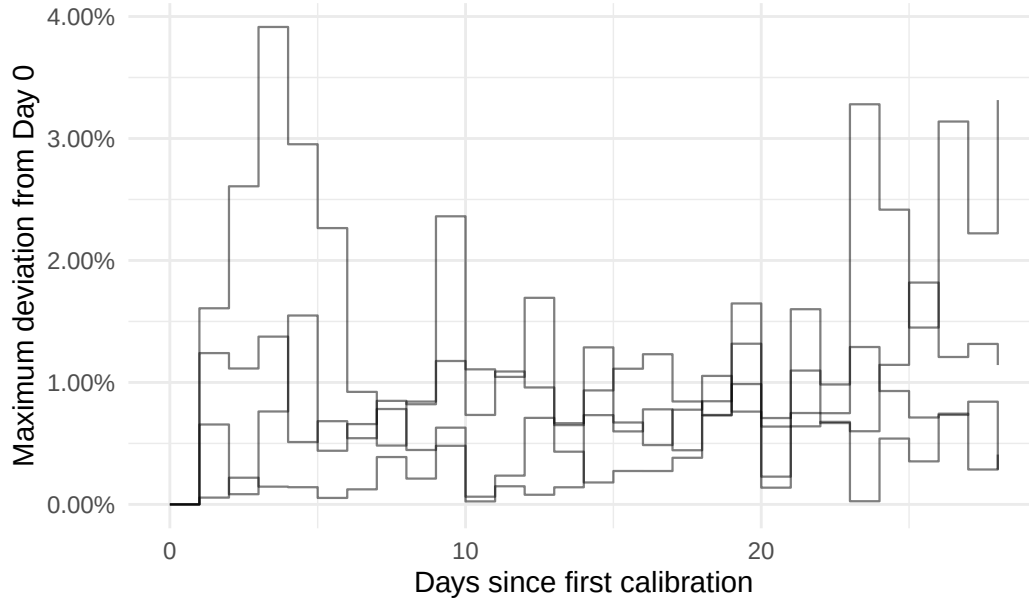
This table lists the United States, Germany, Italy, Singapore, the United Kingdom, Austria, France, Czechia, Spain, Poland, and the Netherlands as being associated with more than 2 anchors that occupy the 90-99.9th percentile consistently. Given the IPv4 and IPv6 shares of the previously identified host network providers in addition to the section 1.2 ASN

geography, this distribution of countries appears to be reasonable. The city and continent distributions are available in the code as well, but are omitted. Indeed, this table stands in as a sanity check along the anchor contextualization discussion.

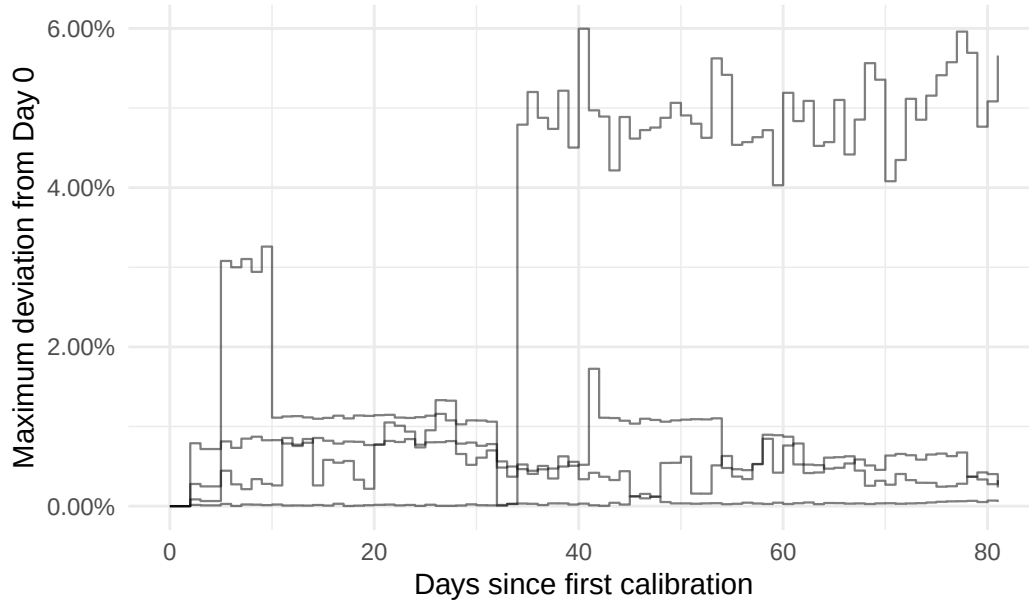
### 3.3 Visualizing Repeated ASN in 90-99.9th Percentile Range

#### 3.3.1 AS14061

2024 Instantaneous Calibration Drift, AS14061 Anchors

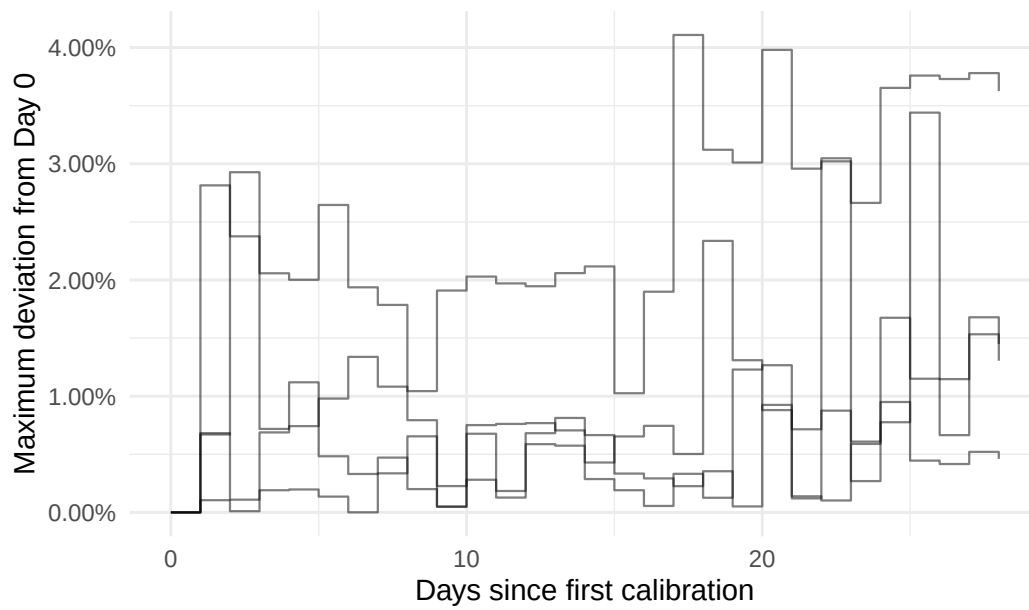


2026 Instantaneous Calibration Drift, AS14061 Anchors

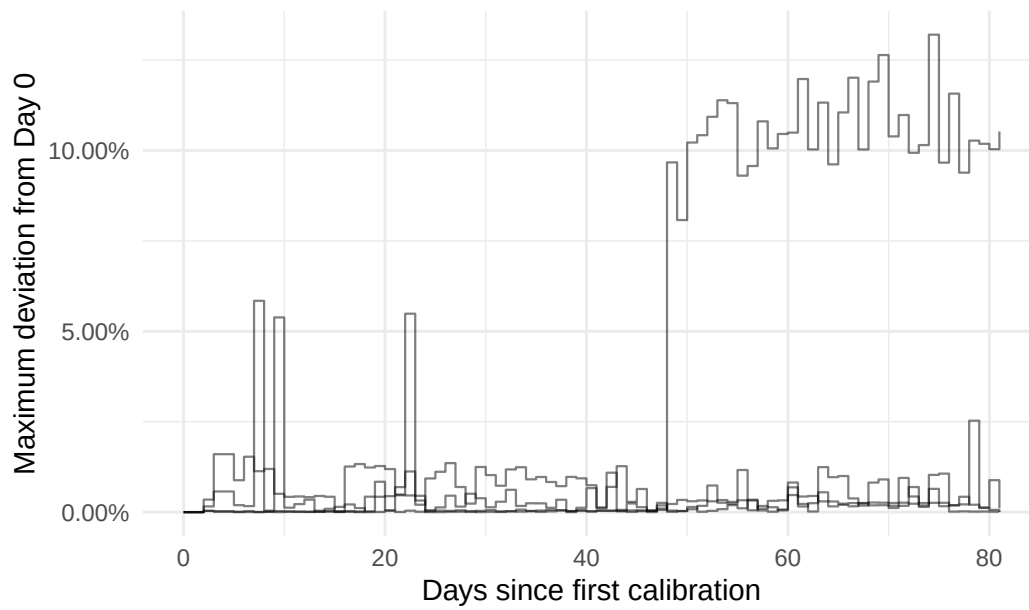


### 3.3.2 AS16276

2024 Instantaneous Calibration Drift, AS16276 Anchors

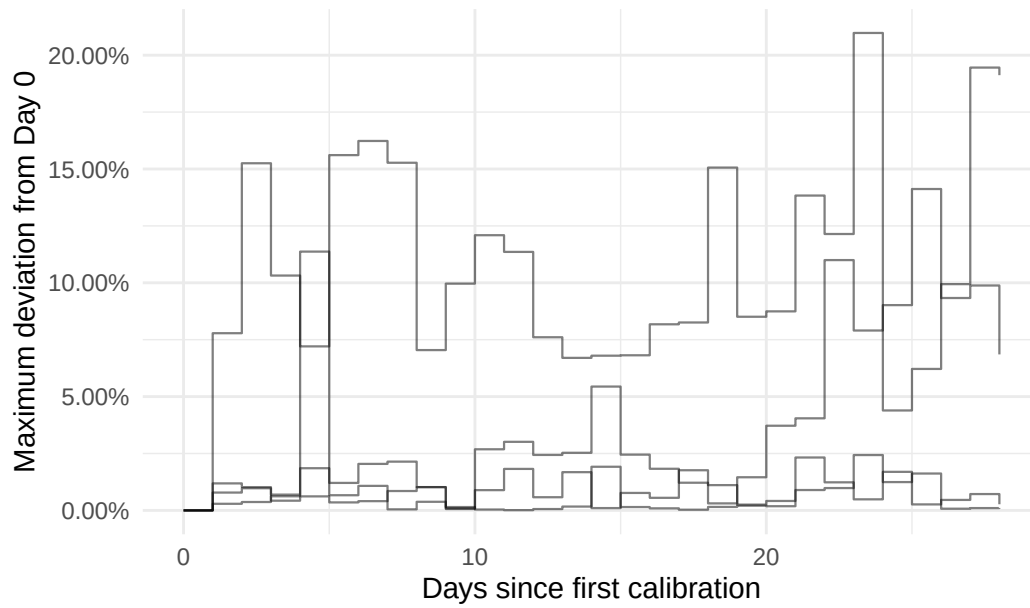


2026 Instantaneous Calibration Drift, AS16276 Anchors

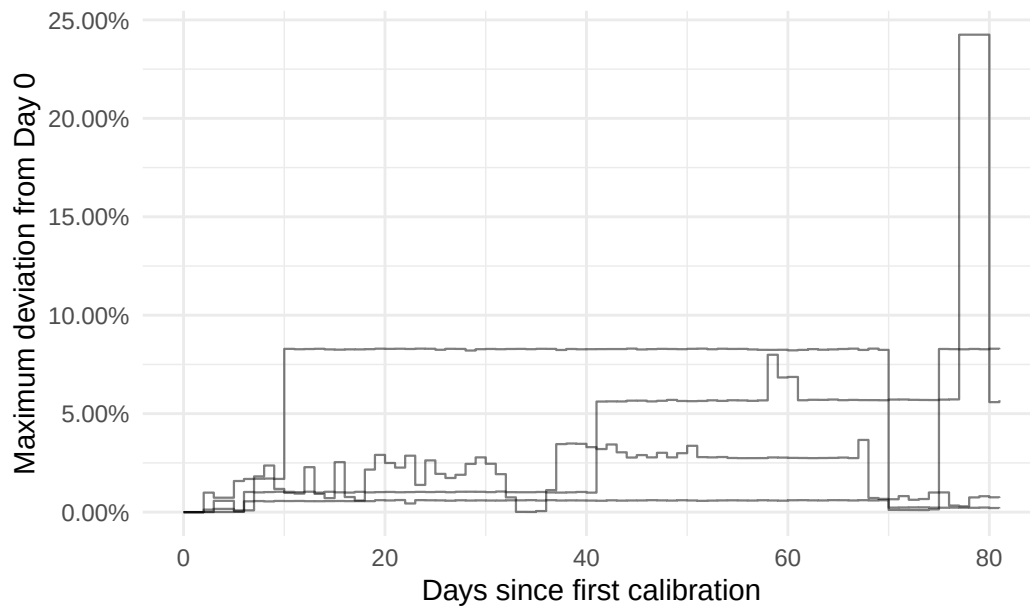


### 3.3.3 AS3491

2024 Instantaneous Calibration Drift, AS3491 Anchors

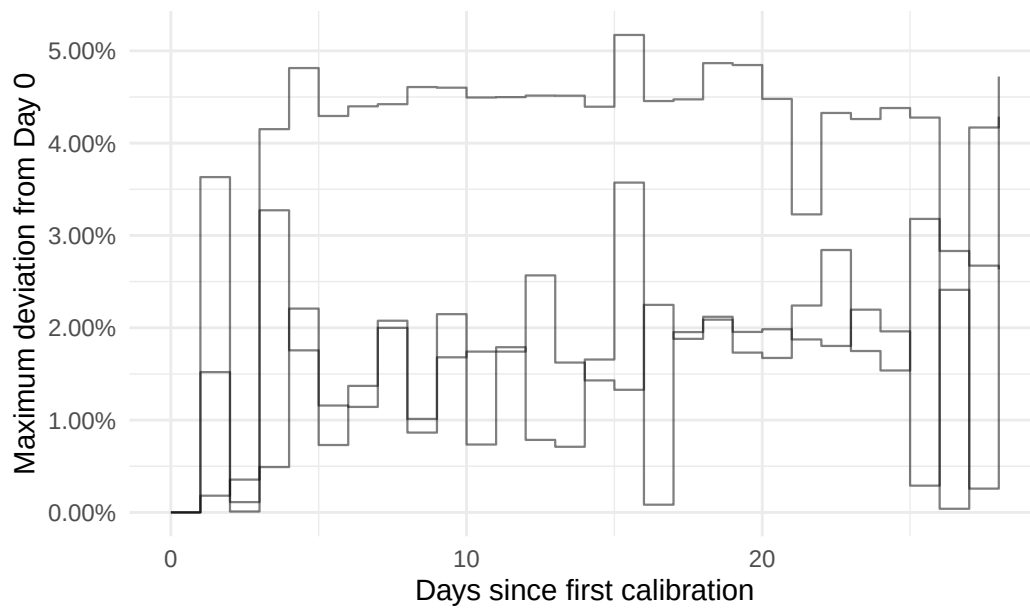


2026 Instantaneous Calibration Drift, AS3491 Anchors

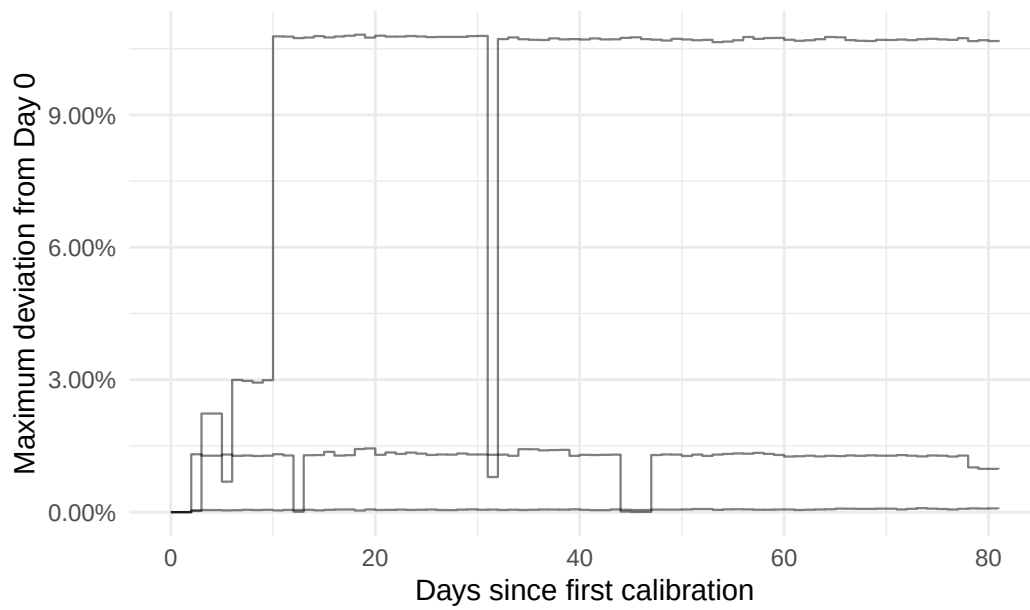


### 3.3.4 AS57695

2024 Instantaneous Calibration Drift, AS57695 Anchors

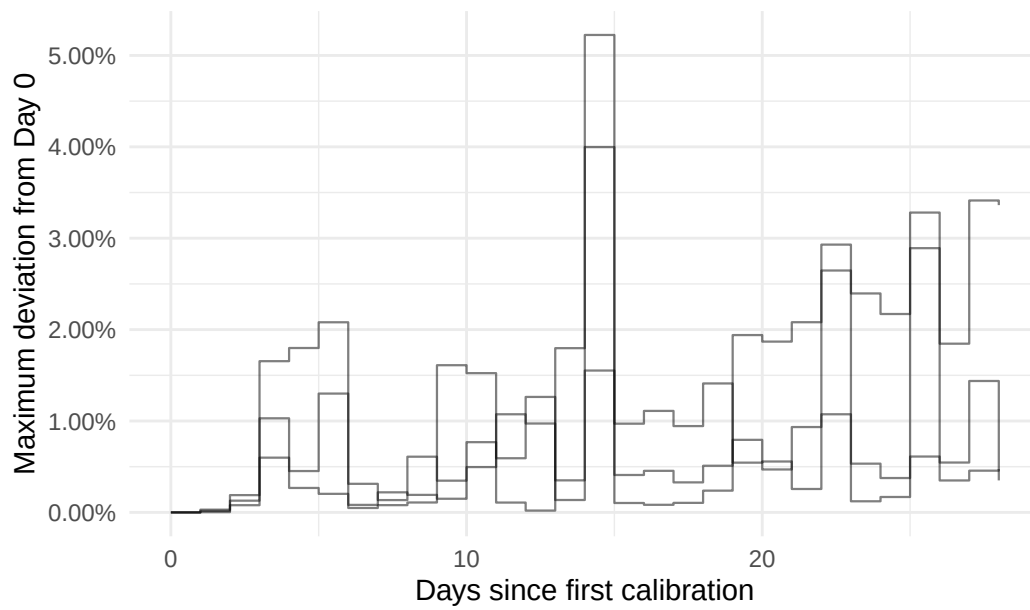


2026 Instantaneous Calibration Drift, AS57695 Anchors

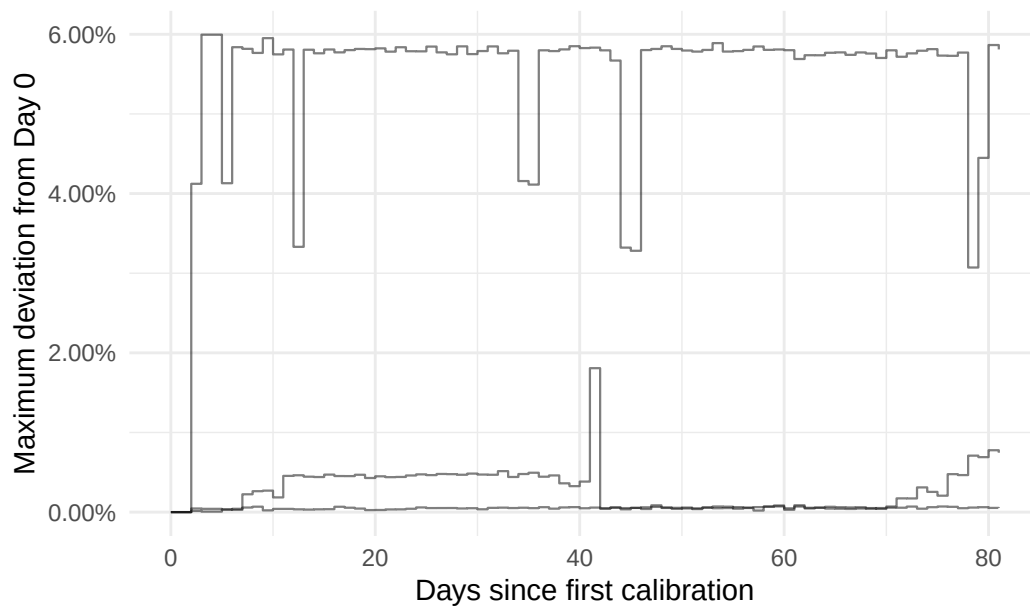


### 3.3.5 AS36236

2024 Instantaneous Calibration Drift, AS36236 Anchors

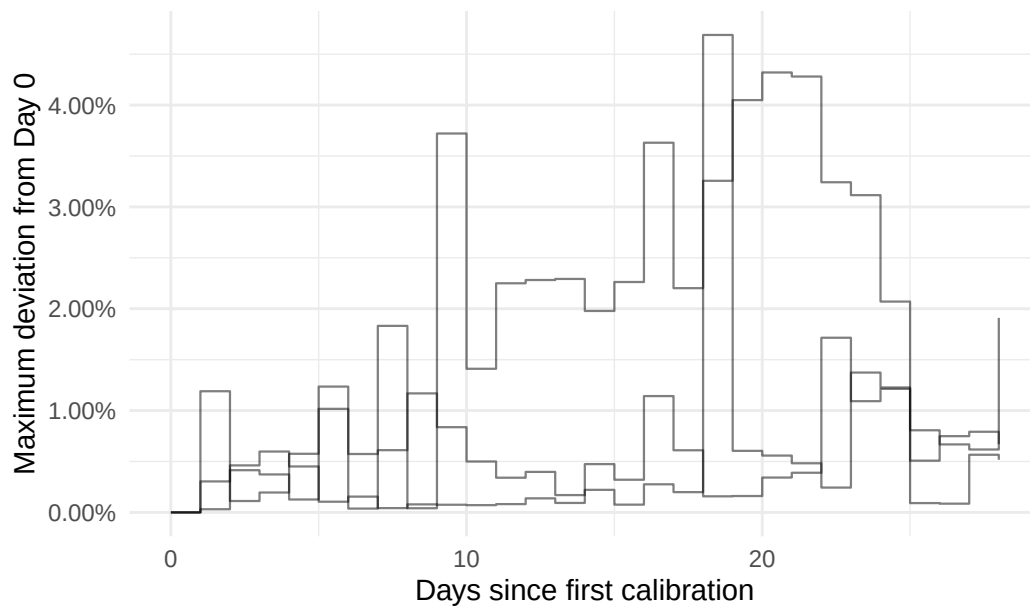


2026 Instantaneous Calibration Drift, AS36236 Anchors

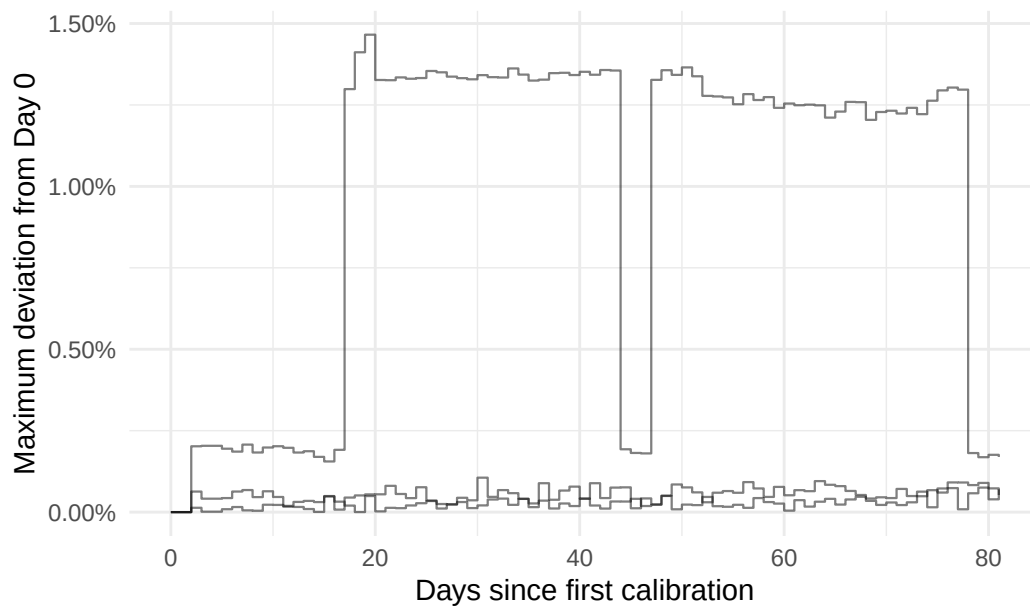


### 3.3.6 AS199524

2024 Instantaneous Calibration Drift, AS199524 Anchors



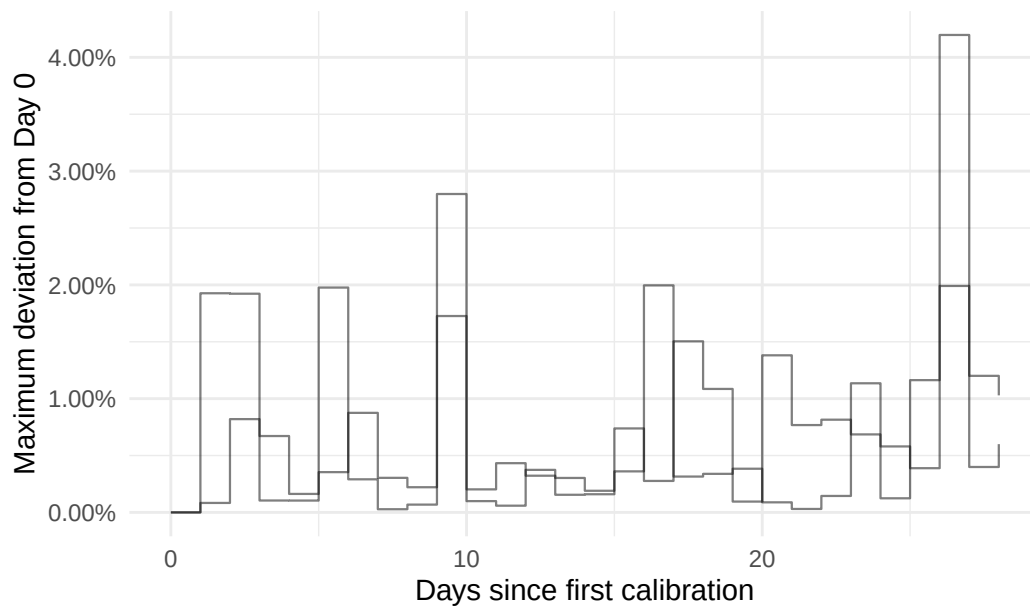
2026 Instantaneous Calibration Drift, AS199524 Anchors



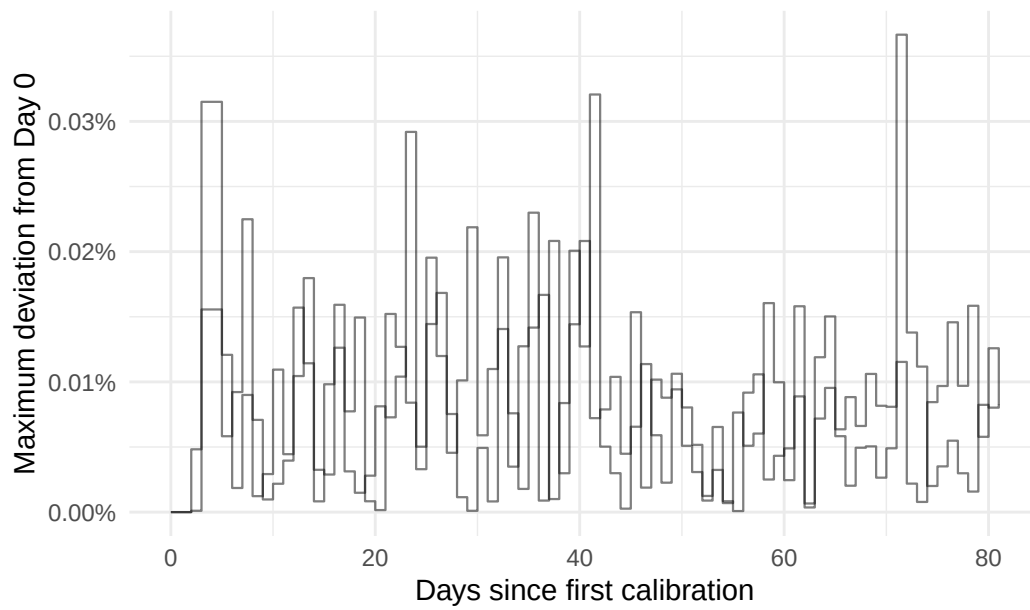


### 3.3.7 AS8473

2024 Instantaneous Calibration Drift, AS8473 Anchors

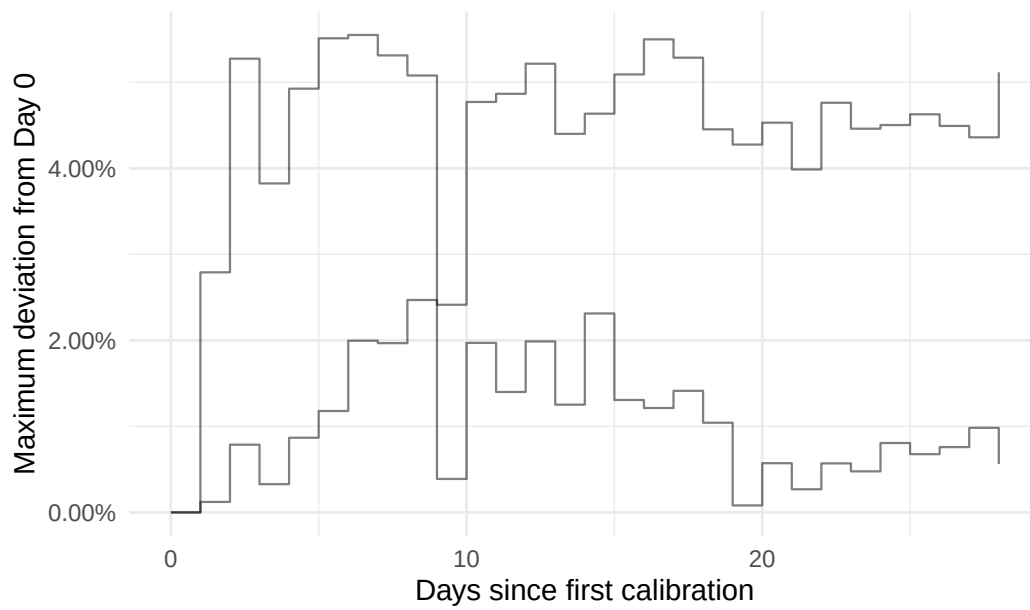


2026 Instantaneous Calibration Drift, AS8473 Anchors

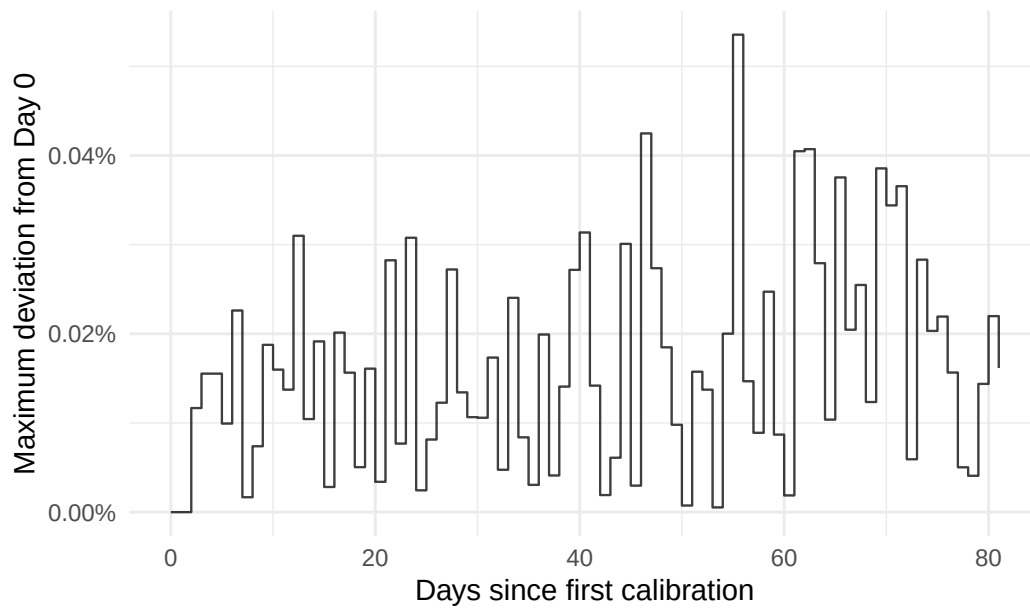


### 3.3.8 AS16347

2024 Instantaneous Calibration Drift, AS16347 Anchors

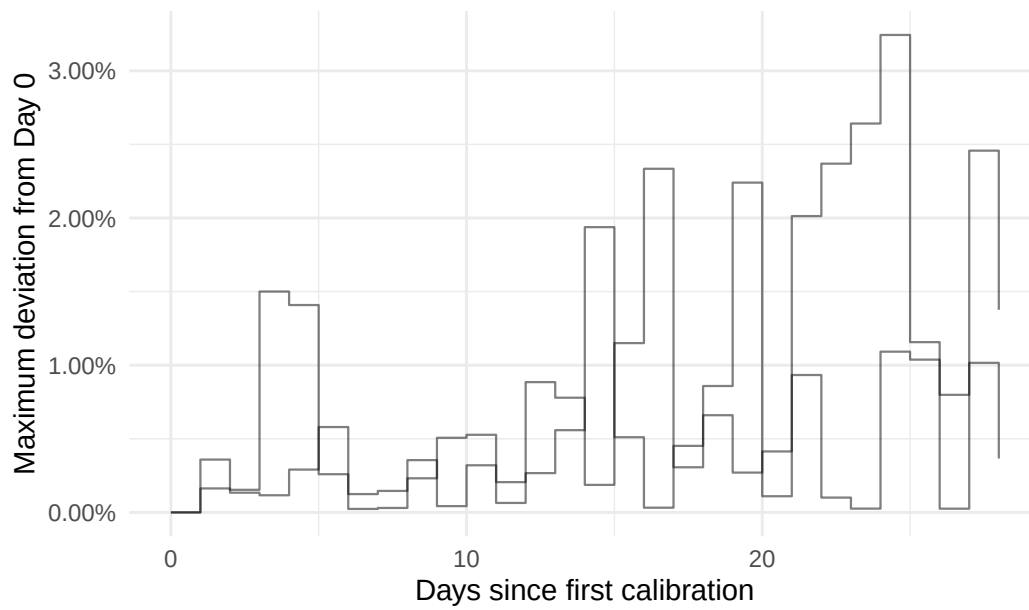


2026 Instantaneous Calibration Drift, AS16347 Anchors

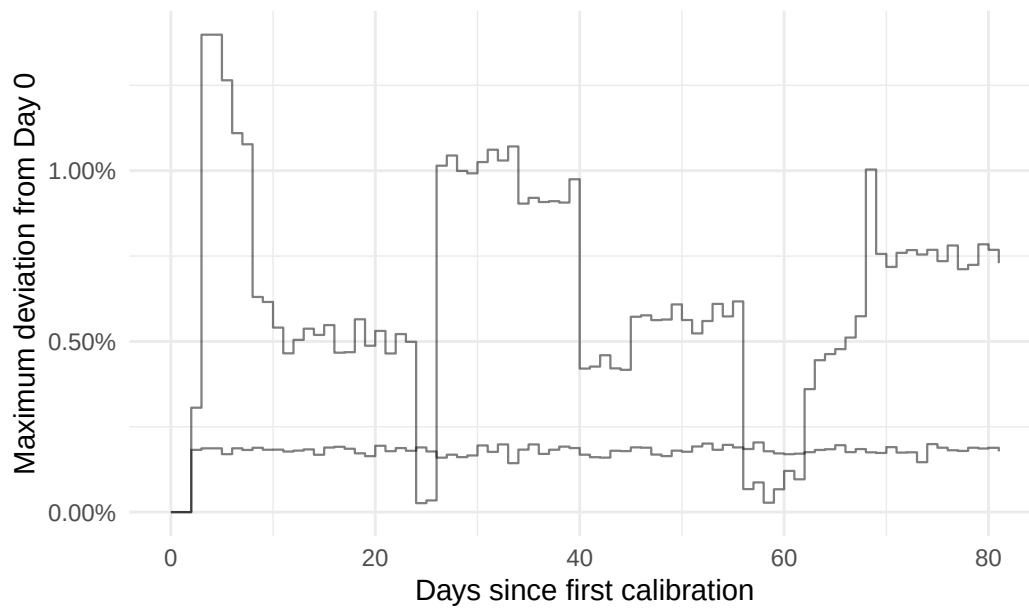


### 3.3.9 AS20473

2024 Instantaneous Calibration Drift, AS20473 Anchors

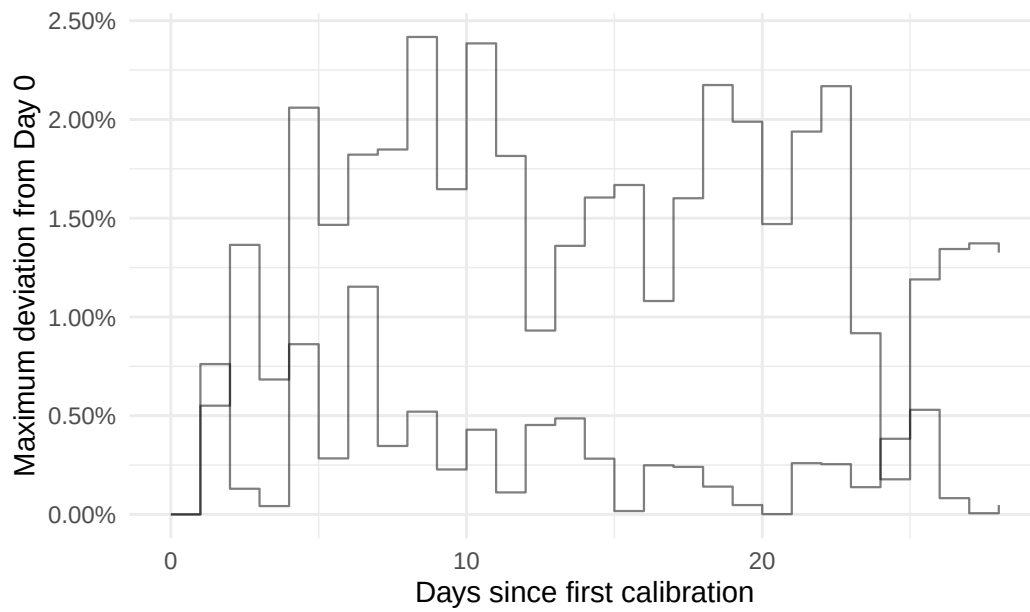


2026 Instantaneous Calibration Drift, AS20473 Anchors

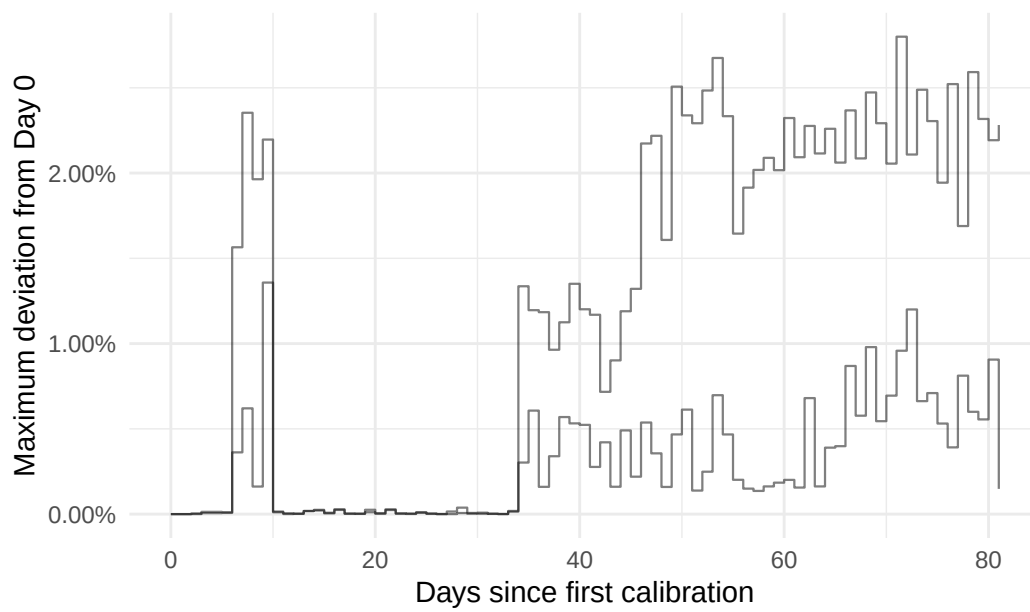


### 3.3.10 AS24940

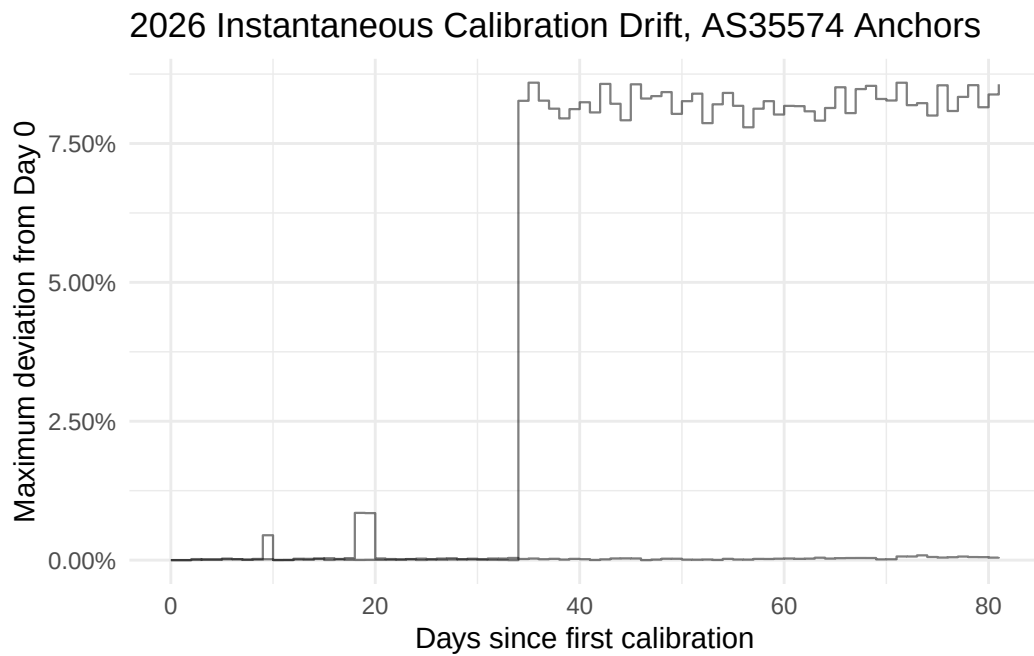
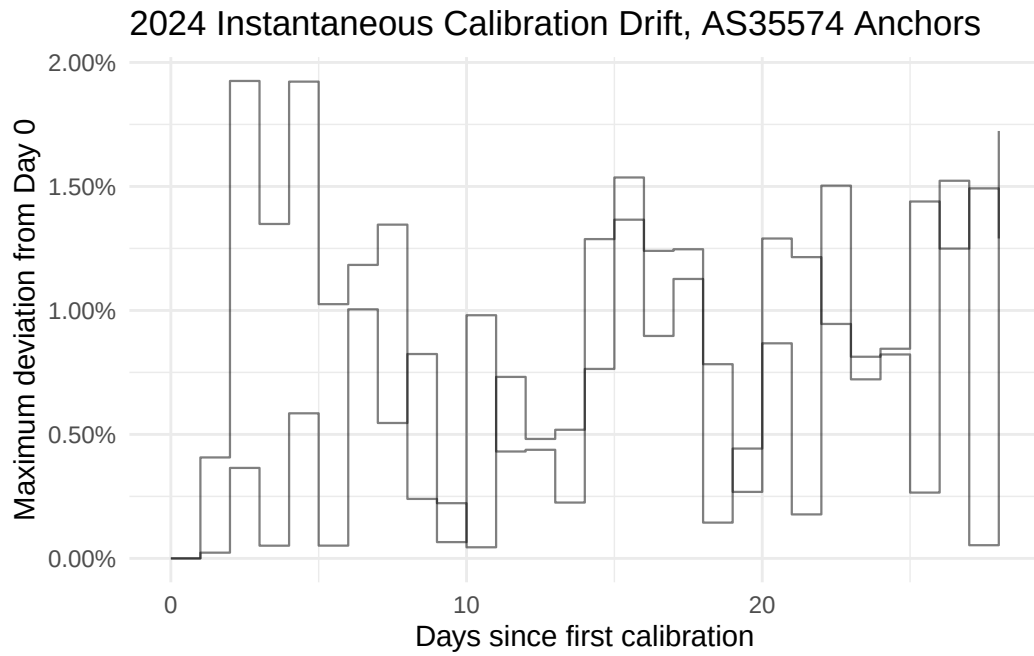
2024 Instantaneous Calibration Drift, AS24940 Anchors



2026 Instantaneous Calibration Drift, AS24940 Anchors

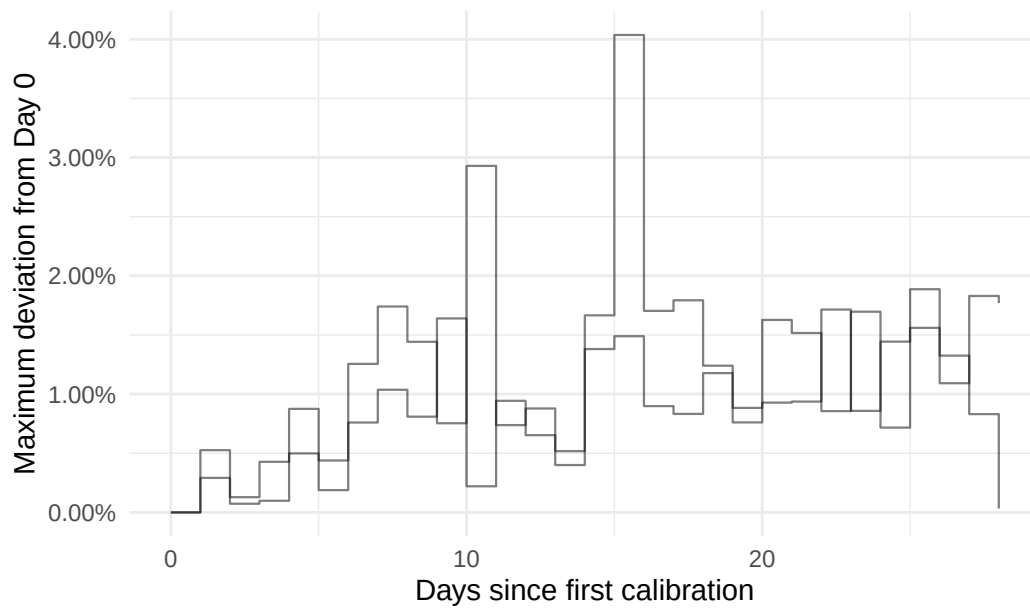


### 3.3.11 AS35574

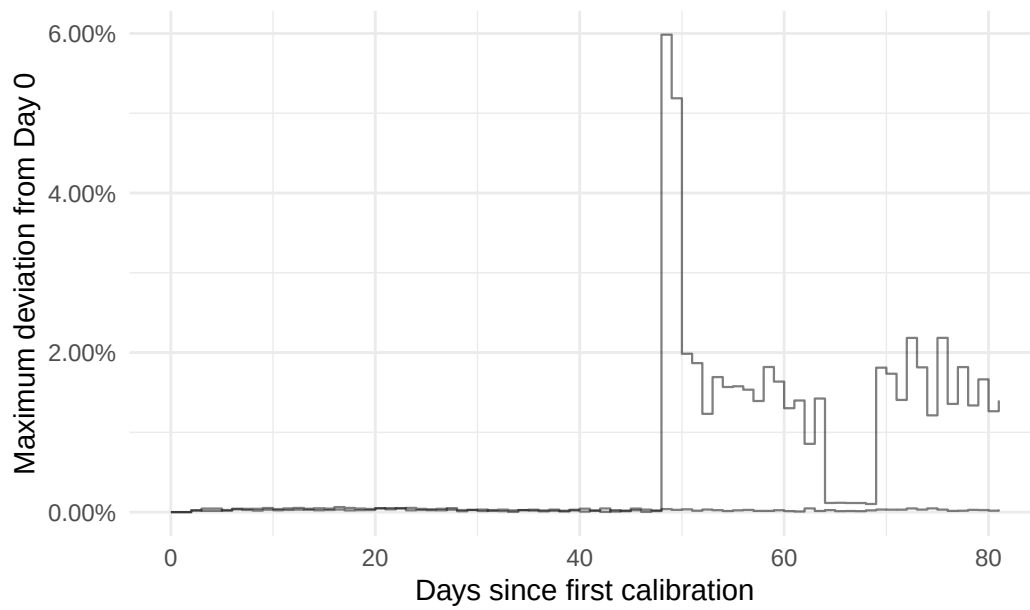


### 3.3.12 AS39923

2024 Instantaneous Calibration Drift, AS39923 Anchors

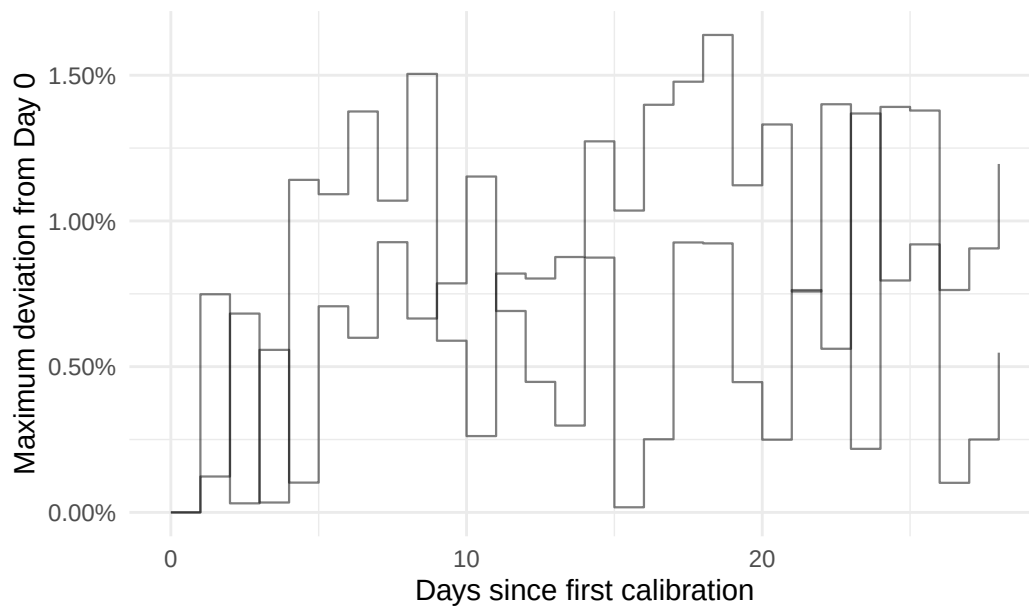


2026 Instantaneous Calibration Drift, AS39923 Anchors

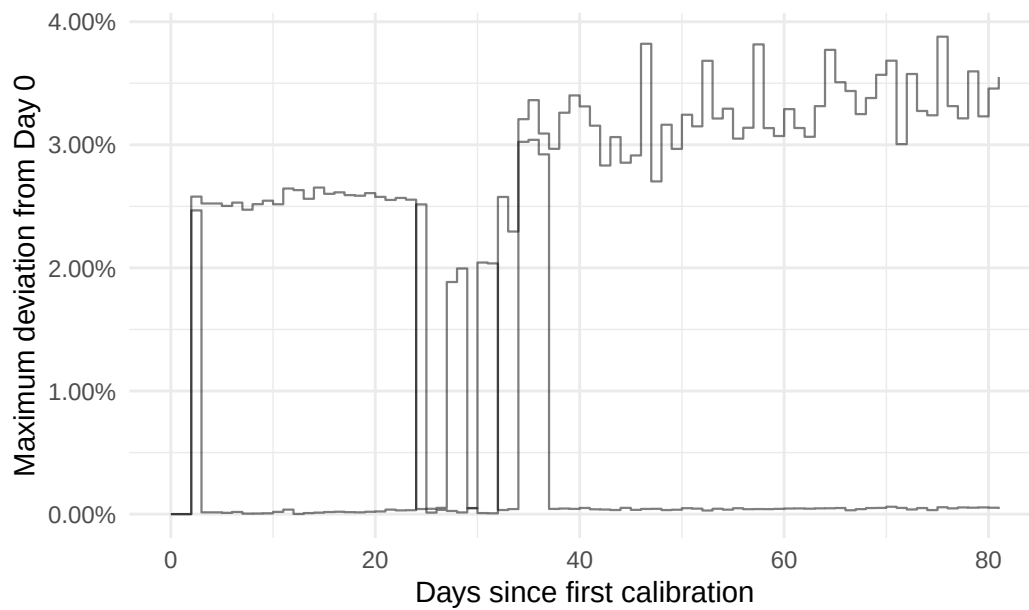


### 3.3.13 AS197540

2024 Instantaneous Calibration Drift, AS197540 Anchors

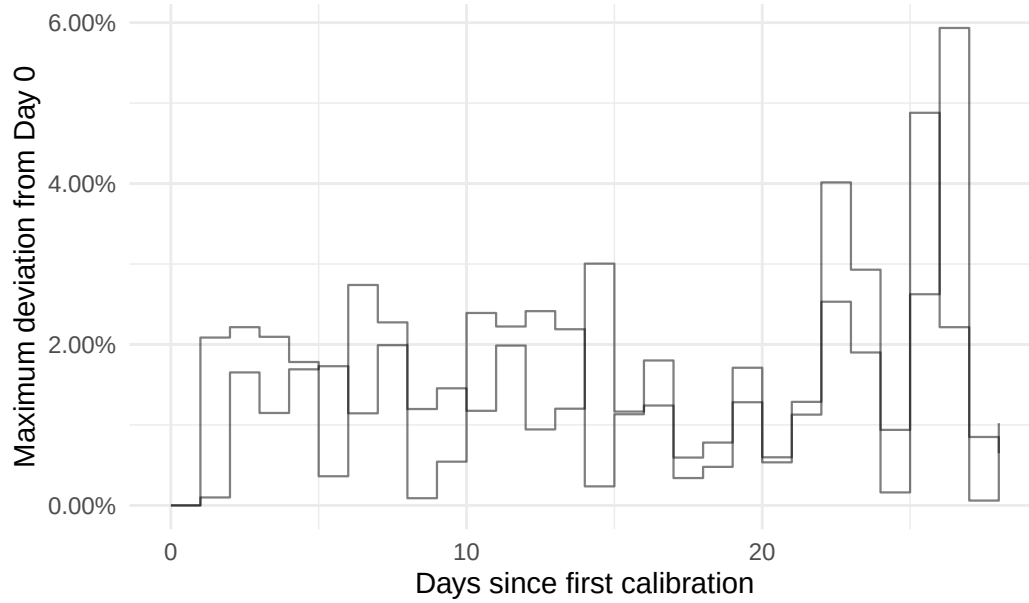


2026 Instantaneous Calibration Drift, AS197540 Anchors

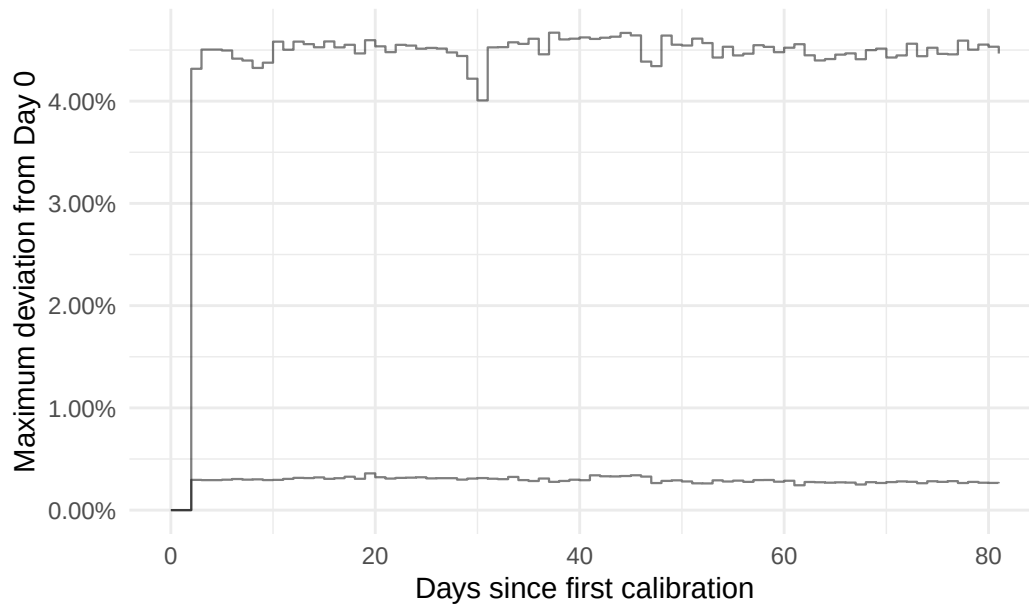


### 3.3.14 AS199610

2024 Instantaneous Calibration Drift, AS199610 Anchors



2026 Instantaneous Calibration Drift, AS199610 Anchors





### 3.4 Flaw with Instantaneous Calibration Drift Visualization: Day 0 as Ground Truth

Once we visualize the individual anchor behaviors associated with repeated ASNs within the 90-999 range, we find once again that all but one of the ASNs maintain instantaneous drift below 15%. These 14 ASNs that appeared more than once in the 90-999 range seemed to be associated with anchor IDs with very conservative, with the smallest drift range being from 0 to 0.04% for AS8473 in 2026. Of the country specific ASNs, AS8473 of Sweden and AS16347 of France had the smallest range of drift maintained, between 0-0.04% and 0-0.05% respectively. AS35574 of Italy and AS39923 of Belgium did not demonstrate particularly small ranges of instantaneous drift, ranging from 0-8% at greatest. It appears that the shift up in instantaneous calibration drift is indeed a factor contributing to the widening of a range that would otherwise be reliably stable due to the nature of its consolidated and locally hosted network. AS3491, however, appears to be responsible for shifting the 99.9th percentile towards 25% in 2026 and above 20% in 2024 once again.

Take a look at the individual anchors associated with the ASN Frequency Table 2. Highlighted are 6 IDs that likely entered the 90-99.9th percentile range due to the strict day 0 ground truth assumption of both the Maximum and Instantaneous Drift visualizations.

Table 13: AS14061 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6968 | 7           | 29       | 4           | 80      | Asia      | IN      | Bangalore |
| 6967 | 1           | 29       | 49          | 80      | Europe    | DE      | Frankfurt |
| 6970 | 1           | 29       | 28          | 80      | Asia      | SG      | Singapore |
| 6971 | 1           | 29       | 4           | 80      | Europe    | GB      | London    |

Table 14: AS16276 Anchors

| id   | in_range_24 | to-tal_24 | in_range_26 | total26 | continent | country | city              |
|------|-------------|-----------|-------------|---------|-----------|---------|-------------------|
| 6800 | 14          | 29        | 1           | 80      | Europe    | PL      | Ozarów Mazowiecki |
| 6727 | 5           | 29        | 5           | 80      | Oceania   | AU      | Sydney            |
| 6696 | 1           | 29        | 35          | 80      | Europe    | FR      | Gravelines        |
| 6726 | 1           | 29        | 17          | 80      | Asia      | SG      | Singapore         |

Table 15: AS3491 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city         |
|------|-------------|----------|-------------|---------|-----------|---------|--------------|
| 7198 | 24          | 29       | 49          | 80      | Asia      | AE      | Dubai        |
| 7078 | 14          | 29       | 70          | 80      | Oceania   | AU      | Sydney       |
| 6779 | 2           | 29       | 1           | 80      | Africa    | ZA      | Johannesburg |

|      |   |    |    |    |      |    |           |
|------|---|----|----|----|------|----|-----------|
| 7010 | 1 | 29 | 33 | 80 | Asia | SG | Singapore |
|------|---|----|----|----|------|----|-----------|

Table 16: AS57695 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city     |
|------|-------------|----------|-------------|---------|---------------|---------|----------|
| 6927 | 26          | 29       | 31          | 80      | North America | US      | Secaucus |
| 6928 | 5           | 29       | 1           | 80      | North America | US      | Reston   |
| 7011 | 3           | 29       | 56          | 80      | North America | US      | Seattle  |

Table 17: AS36236 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city      |
|------|-------------|----------|-------------|---------|---------------|---------|-----------|
| 6379 | 5           | 29       | 1           | 80      | North America | US      | Durham    |
| 6408 | 2           | 29       | 79          | 80      | North America | US      | Phoenix   |
| 6487 | 2           | 29       | 1           | 80      | Asia          | SG      | Singapore |

Table 18: AS199524 Anchors

| id   | in_range_24 | to-<br>tal_24 | in_range_26 | total26 | continent        | country | city     |
|------|-------------|---------------|-------------|---------|------------------|---------|----------|
| 6668 | 8           | 29            | 1           | 80      | Europe           | ES      | Madrid   |
| 6663 | 2           | 29            | 1           | 80      | Europe           | IT      | Milano   |
| 6644 | 1           | 29            | 18          | 80      | North<br>America | US      | Secaucus |

Table 19: AS8473 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 7127 | 3           | 29       | 1           | 80      | Europe    | SE      | Malmö     |
| 7109 | 2           | 29       | 1           | 80      | Europe    | SE      | Stockholm |

Table 20: AS16347 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6443 | 28          | 29       | 1           | 80      | Europe    | FR      | Maxeville |
| 7048 | 2           | 29       | 1           | 80      | Europe    | FR      | Nancy     |

Table 21: AS20473 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 7037 | 2           | 29       | 1           | 80      | Europe    | SE      | Stockholm |
| 6506 | 1           | 29       | 8           | 80      | Asia      | SG      | Singapore |

Table 22: AS24940 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6693 | 3           | 29       | 2           | 80      | Europe    | DE      | Nuremberg |
| 6332 | 1           | 29       | 5           | 80      | Europe    | DE      | Nuremberg |

Table 23: AS35574 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city    |
|------|-------------|----------|-------------|---------|-----------|---------|---------|
| 6450 | 2           | 29       | 1           | 80      | Europe    | IT      | Rome    |
| 6435 | 1           | 29       | 49          | 80      | Europe    | IT      | Rozzano |

Table 24: AS39923 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city     |
|------|-------------|----------|-------------|---------|-----------|---------|----------|
| 6635 | 3           | 29       | 1           | 80      | Europe    | BE      | Leuven   |
| 6737 | 1           | 29       | 3           | 80      | Europe    | BE      | Zaventem |

Table 25: AS197540 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6581 | 1           | 29       | 31          | 80      | Europe    | DE      | Nuremberg |
| 6799 | 1           | 29       | 29          | 80      | Europe    | DE      | Nürnberg  |

Table 26: AS199610 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city        |
|------|-------------|----------|-------------|---------|---------------|---------|-------------|
| 6649 | 9           | 29       | 80          | 80      | North America | US      | Los Angeles |

### 3.5 Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range

We maintain AS3941 as associated with outlier drift behavior among the repeated ASNs. Below are the IDs associated with unique ASNs that entered the 90-999 range; we will then take away the likely misplaced IDs, highlighted in yellow, due to the day 0 limitation.

Table 27: Anchors Associated with Unique ASNs

| asn    | id   | in_range_24 | total_24 | in_range_26 | total26 | country | city              |
|--------|------|-------------|----------|-------------|---------|---------|-------------------|
| 719    | 6251 | 1           | 29       | 2           | 80      | FI      | Tampere           |
| 203329 | 6313 | 1           | 29       | 32          | 80      | DE      | Frankfurt am Main |
| 48943  | 6325 | 1           | 29       | 49          | 80      | AT      | Vienna            |
| 2108   | 6327 | 1           | 29       | 30          | 80      | HR      | Zagreb            |
| 62416  | 6328 | 1           | 29       | 73          | 80      | PT      | Lisbon            |
| 1853   | 6334 | 1           | 29       | 13          | 80      | AT      | Vienna            |
| 2857   | 6336 | 1           | 29       | 5           | 80      | DE      | Mainz             |
| 41000  | 6423 | 1           | 29       | 2           | 80      | GB      | Manchester        |
| 7247   | 6482 | 1           | 29       | 2           | 80      | US      | Fremont           |
| 31034  | 6509 | 1           | 29       | 49          | 80      | IT      | Arezzo            |
| 201126 | 6516 | 1           | 29       | 2           | 80      | GB      | London            |
| 39642  | 6545 | 1           | 29       | 6           | 80      | DK      | Kolding           |
| 41095  | 6568 | 1           | 29       | 4           | 80      | SG      | Singapore         |
| 39617  | 6674 | 1           | 29       | 3           | 80      | GB      | Balderton         |
| 24482  | 6676 | 1           | 29       | 25          | 80      | SG      | Singapore         |
| 786    | 6695 | 1           | 29       | 35          | 80      | GB      | Didcot            |
| 25376  | 6716 | 1           | 29       | 2           | 80      | GB      | Bolton            |
| 62310  | 6751 | 1           | 29       | 5           | 80      | DE      | Berlin            |
| 41327  | 6762 | 1           | 29       | 46          | 80      | IT      | Palermo           |
| 199399 | 6841 | 1           | 29       | 2           | 80      | GR      | Athens            |
| 21320  | 6874 | 1           | 29       | 3           | 80      | FR      | Paris             |
| 62513  | 6888 | 1           | 29       | 5           | 80      | CA      | Toronto           |
| 63949  | 6918 | 1           | 29       | 6           | 80      | SG      | Singapore         |
| 39835  | 6922 | 1           | 29       | 2           | 80      | DE      | Göttingen         |
| 1213   | 6965 | 1           | 29       | 2           | 80      | IE      | Dublin            |
| 39636  | 7003 | 1           | 29       | 27          | 80      | NL      | Arnhem            |
| 30971  | 7008 | 1           | 29       | 2           | 80      | AT      | Vienna            |
| 13101  | 7012 | 1           | 29       | 4           | 80      | DE      | Kiel              |
| 38001  | 7091 | 1           | 29       | 4           | 80      | SG      | Singapore         |
| 56381  | 7097 | 1           | 29       | 53          | 80      | DE      | Frankfurt am Main |
| 33920  | 7125 | 1           | 29       | 32          | 80      | GB      | Leeds             |
| 29423  | 7141 | 1           | 29       | 28          | 80      | DE      | Frankfurt         |

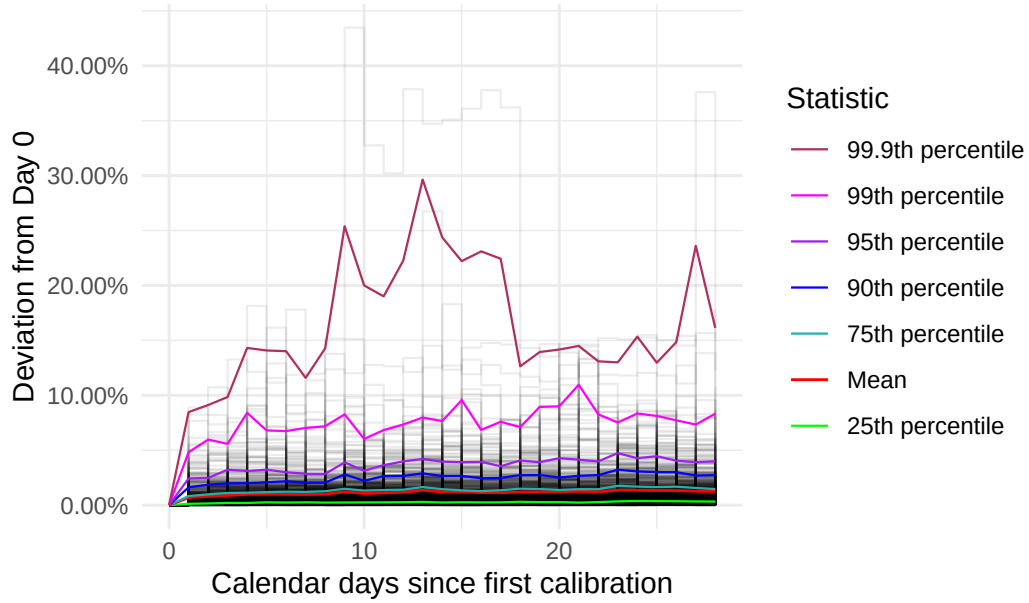
|        |      |   |    |    |    |    |                          |
|--------|------|---|----|----|----|----|--------------------------|
| 50030  | 7156 | 1 | 29 | 5  | 80 | DE | Berlin                   |
| 39063  | 7169 | 1 | 29 | 49 | 80 | DE | Frankfurt                |
| 60841  | 7201 | 1 | 29 | 4  | 80 | NL | Amsterdam                |
| 35492  | 7216 | 1 | 29 | 31 | 80 | AT | Vienna                   |
| 54316  | 7224 | 1 | 29 | 9  | 80 | US | Greenwich                |
| 29287  | 7225 | 1 | 29 | 47 | 80 | AT | Vienna                   |
| 23428  | 7238 | 1 | 29 | 69 | 80 | US | Kansas City              |
| 14907  | 7261 | 1 | 29 | 2  | 80 | NL | Amsterdam                |
| 216444 | 7272 | 1 | 29 | 13 | 80 | SG | Singapore                |
| 32167  | 7292 | 1 | 29 | 10 | 80 | HK | Hong Kong                |
| 34428  | 7295 | 1 | 29 | 47 | 80 | IT | Carini                   |
| 50178  | 6424 | 2 | 29 | 1  | 80 | IT | Meran                    |
| 60396  | 6441 | 2 | 29 | 1  | 80 | SE | Lund                     |
| 39392  | 6544 | 2 | 29 | 1  | 80 | CZ | Prague                   |
| 199188 | 6559 | 2 | 29 | 1  | 80 | GB | London                   |
| 3320   | 6724 | 2 | 29 | 1  | 80 | DE | Frankfurt am Main        |
| 46997  | 6794 | 2 | 29 | 78 | 80 | US | Seattle                  |
| 23483  | 7052 | 2 | 29 | 1  | 80 | US | Redding                  |
| 50599  | 7064 | 2 | 29 | 1  | 80 | PL | Torun                    |
| 5459   | 7081 | 2 | 29 | 1  | 80 | GB | London                   |
| 31510  | 7083 | 2 | 29 | 1  | 80 | AT | Innsbruck                |
| 47295  | 7096 | 2 | 29 | 1  | 80 | DE | Bautzen                  |
| 59678  | 7193 | 2 | 29 | 72 | 80 | US | Piscataway               |
| 12110  | 7199 | 2 | 29 | 25 | 80 | US | Tampa                    |
| 25248  | 7211 | 2 | 29 | 2  | 80 | CZ | Prague                   |
| 13303  | 7213 | 2 | 29 | 1  | 80 | RS | Nis                      |
| 41931  | 7249 | 2 | 29 | 1  | 80 | PL | Bialystok                |
| 6724   | 6331 | 3 | 29 | 1  | 80 | DE | Berlin                   |
| 39878  | 6360 | 3 | 29 | 49 | 80 | AT | Salzburg                 |
| 204092 | 6494 | 3 | 29 | 1  | 80 | FR | Rennes                   |
| 59715  | 6496 | 3 | 29 | 2  | 80 | IT | San Benedetto del Tronto |
| 24971  | 6809 | 3 | 29 | 3  | 80 | CZ | Brno                     |
| 29670  | 6842 | 3 | 29 | 1  | 80 | DE | Berlin                   |
| 38064  | 6904 | 3 | 29 | 26 | 80 | NZ | Auckland                 |
| 6881   | 7024 | 3 | 29 | 1  | 80 | CZ | Prague                   |
| 57692  | 7053 | 3 | 29 | 1  | 80 | FI | Helsinki                 |
| 267    | 7105 | 3 | 29 | 1  | 80 | US | Southfield               |
| 36444  | 6389 | 4 | 29 | 1  | 80 | US | Detroit                  |
| 48362  | 6476 | 4 | 29 | 1  | 80 | AT | Feldkirch                |
| 202297 | 6547 | 4 | 29 | 1  | 80 | CZ | Prague                   |
| 28725  | 6577 | 4 | 29 | 1  | 80 | CZ | Prague                   |
| 42612  | 7017 | 4 | 29 | 2  | 80 | ES | Madrid                   |
| 395823 | 7043 | 4 | 29 | 79 | 80 | US | Seattle                  |
| 21034  | 7110 | 4 | 29 | 5  | 80 | IT | Pescara                  |

|        |      |    |    |    |    |    |                   |
|--------|------|----|----|----|----|----|-------------------|
| 40581  | 7087 | 5  | 29 | 1  | 80 | US | Fayetteville      |
| 1280   | 7253 | 5  | 29 | 1  | 80 | US | Palo Alto         |
| 200187 | 7265 | 5  | 29 | 2  | 80 | DE | Frankfurt am Main |
| 15826  | 7282 | 5  | 29 | 1  | 80 | FR | Toulouse          |
| 29169  | 6363 | 6  | 29 | 1  | 80 | FR | Saint-Denis       |
| 3292   | 6952 | 7  | 29 | 1  | 80 | DK | Højbjerg          |
| 25192  | 7223 | 7  | 29 | 1  | 80 | CZ | Prague            |
| 15954  | 6604 | 9  | 29 | 1  | 80 | ES | Madrid            |
| 24840  | 7196 | 9  | 29 | 49 | 80 | DE | Frankfurt         |
| 39069  | 6687 | 10 | 29 | 1  | 80 | ES | Murcia            |
| 209097 | 6736 | 12 | 29 | 1  | 80 | FR | Trevoux           |
| 15802  | 6962 | 12 | 29 | 1  | 80 | AE | Dubai             |
| 15699  | 7086 | 12 | 29 | 2  | 80 | ES | Alcalá de Henares |
| 12008  | 6954 | 13 | 29 | 32 | 80 | IN | Mumbai            |
| 202196 | 6639 | 14 | 29 | 1  | 80 | HK | Hong Kong         |
| 201333 | 6675 | 14 | 29 | 1  | 80 | IT | Piacenza          |
| 7752   | 7221 | 17 | 29 | 77 | 80 | US | Seattle           |
| 53698  | 7159 | 21 | 29 | 78 | 80 | US | Phoenix           |
| 206186 | 6323 | 25 | 29 | 1  | 80 | PL | Katowice          |

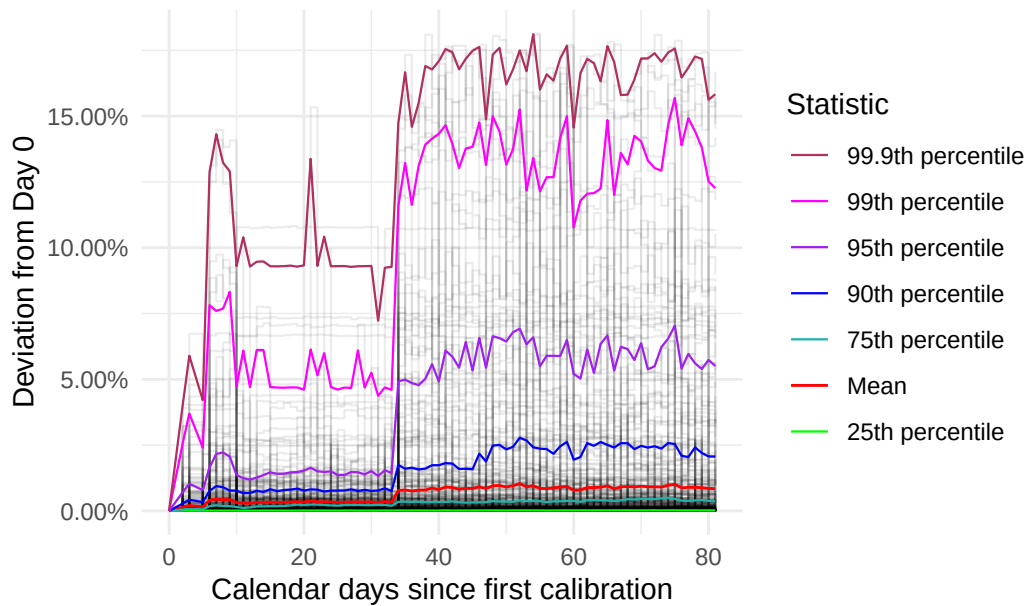
### 3.6 Instantaneous Drift

(after day 0 adjustment of the IDs present in 90-99.9th percentile in Instantaneous Drift, Section 3)

2024 Instantaneous Calibration Drift



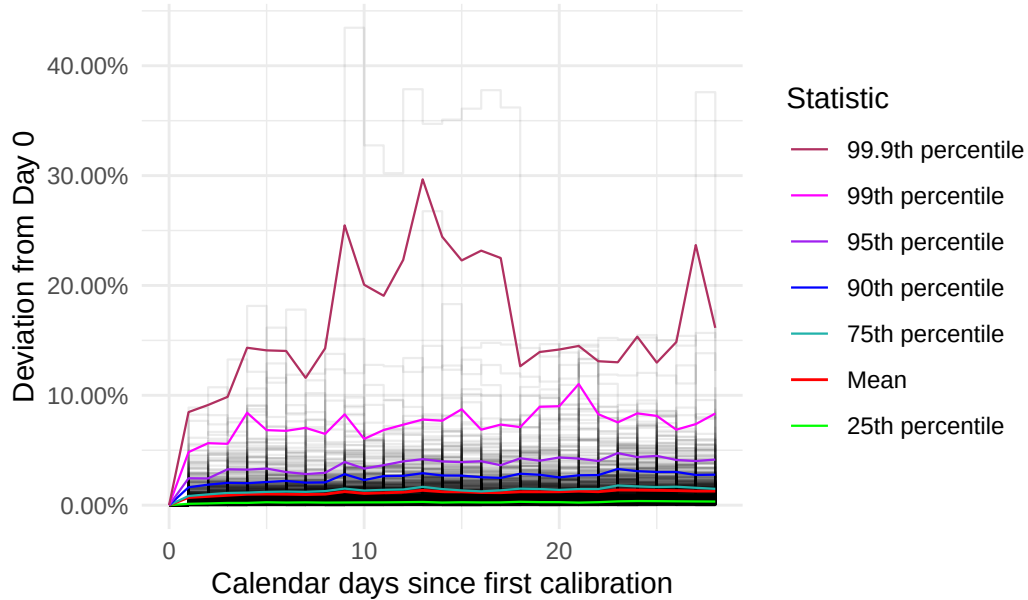
2026 Instantaneous Calibration Drift



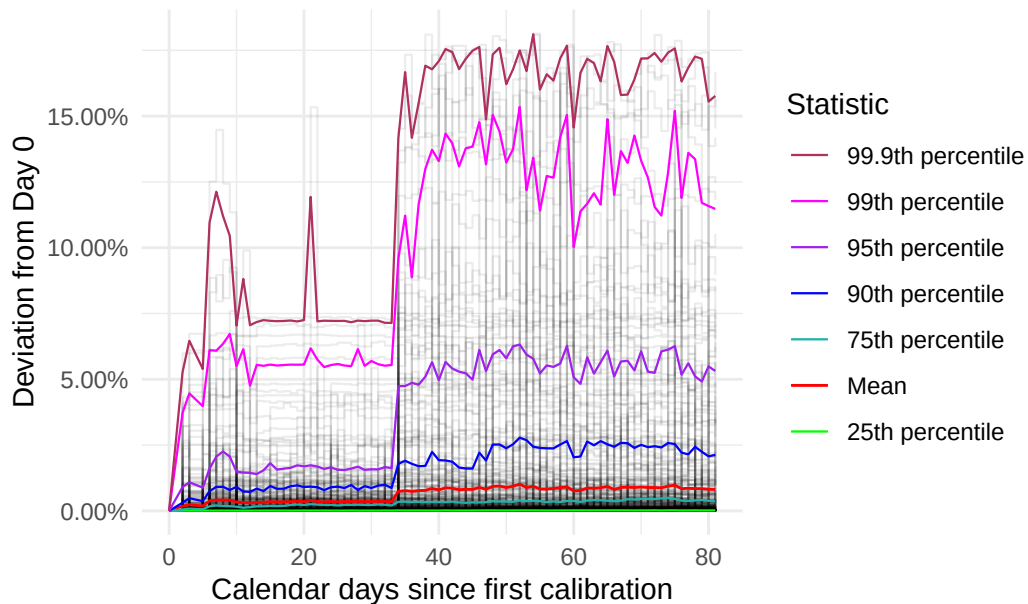
### 3.6.1 Instantaneous Drift

(after day 0 adjustment of the IDs present in 90-99.9th percentile in Maximum Drift, Section 1)

2024 Instantaneous Calibration Drift



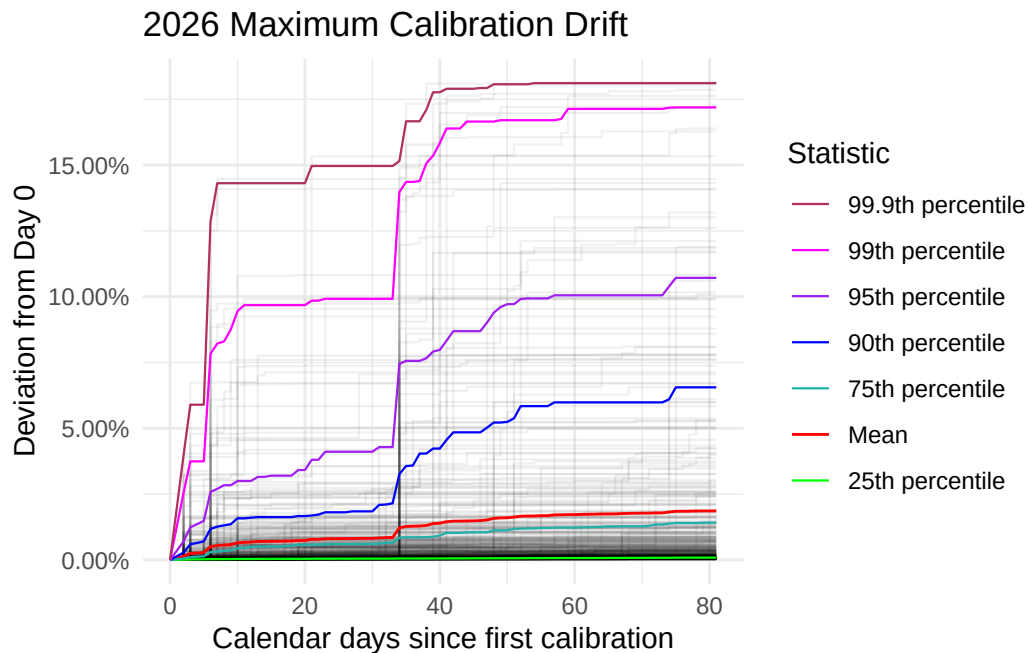
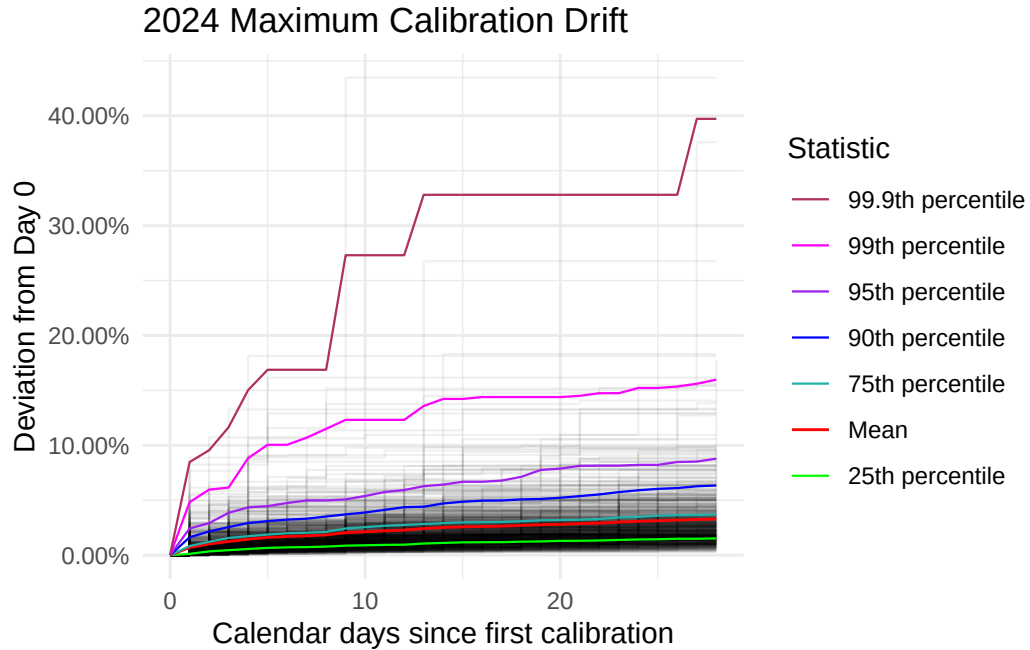
2026 Instantaneous Calibration Drift





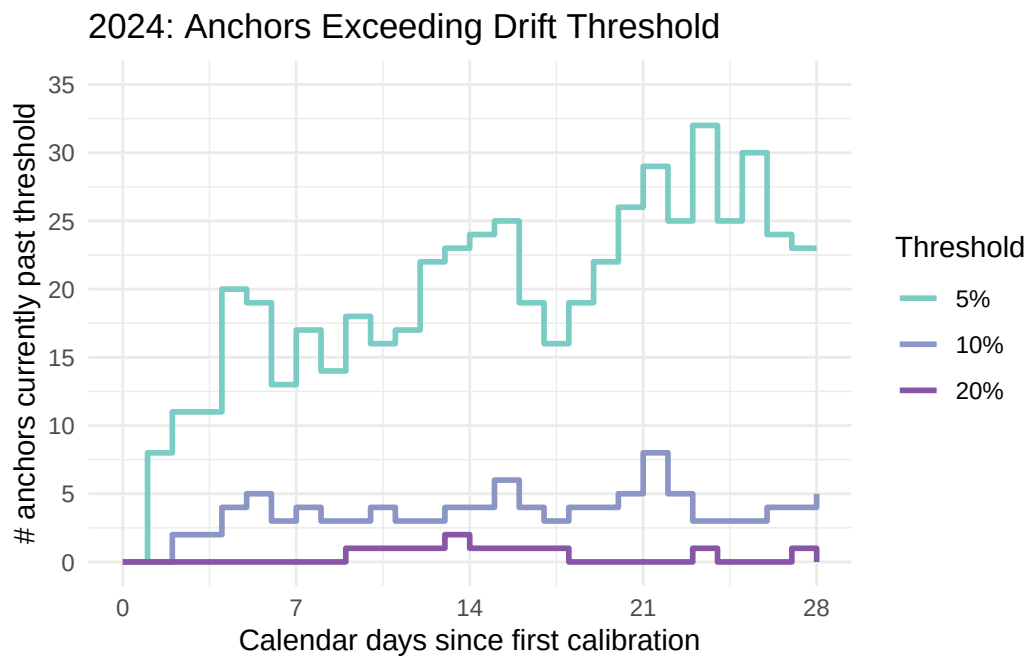
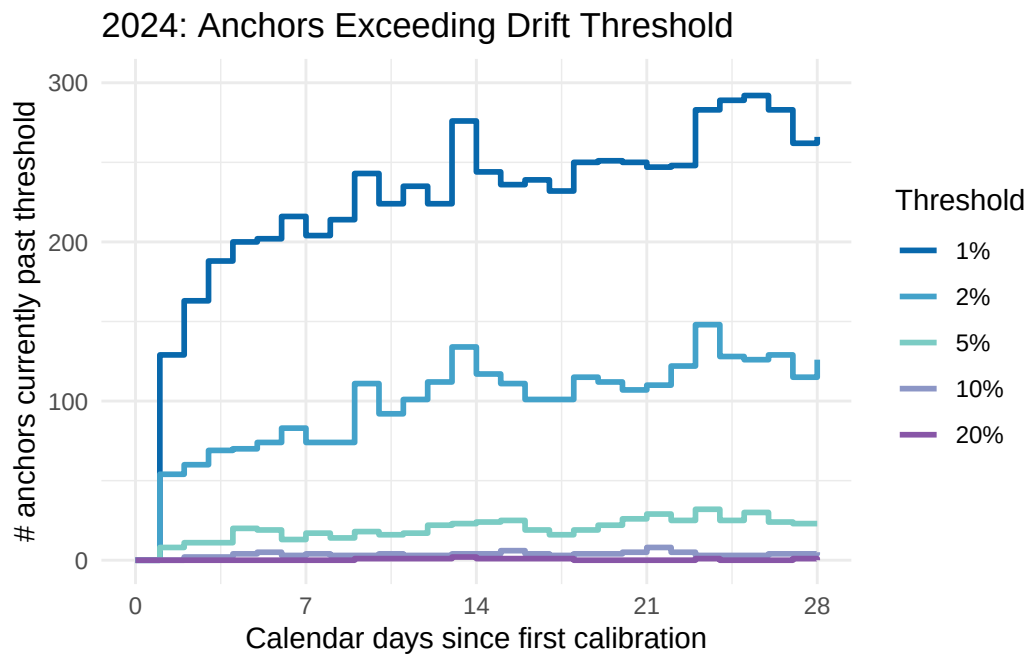
### 3.6.2 Maximum Drift

(after day 0 adjustment of the IDs present in 90-99.9th percentile in Instantaneous Drift, Section 1)

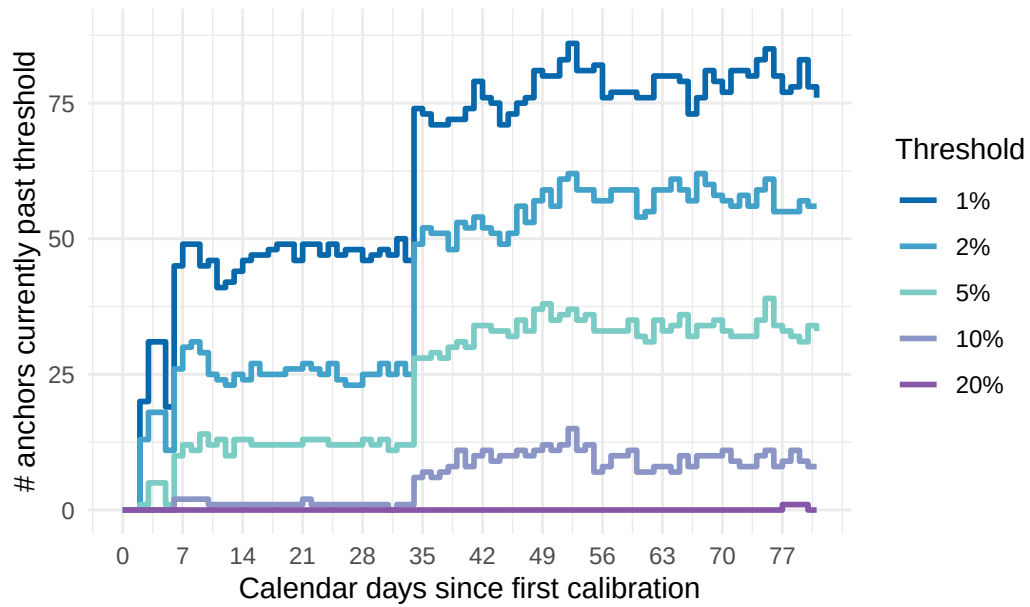


### 3.7 Discussion on Adjusted Instantaneous Drift Visualizations

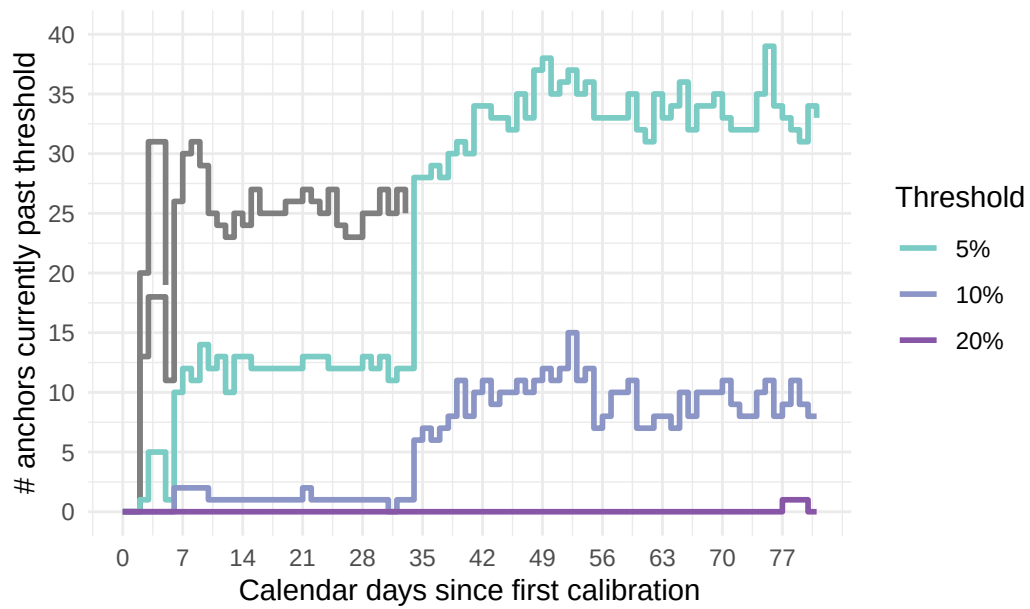
## 4. Instantaneous Drift Threshold



2026: Anchors Exceeding Drift Threshold



2026: Anchors Exceeding Drift Threshold



## 4.1 Observations/Discussion

add mean or median values for each threshold line, note value

EDIT THIS ->The Maximum Drift Threshold uses percentage thresholds to illustrate the rate at which the anchors exceed 1%, 2% 5%, 10%, and 20% drift. By day 7, nearly 400 of 659 anchors in the 2024 data have exceeded 1% drift, and around 200 exceed 2% and 55 exceed 5% by day 9 and 14 respectively. In order to improve interpretability, I have adjusted the y axis for examining the 10% and 20% specific thresholds trends. In this figure, we see that about 20 of 659 anchors exceeded 10% drift, and 4 of 659 exceeded 20% by the end of the recorded period. This visualization is especially convenient for observing the rate at which the anchors collectively exceed each threshold.

For the 2026 data, we use two figures to visualize our threshold details carefully once more. The drift trends for the 2026 data appear to be more conservative than the 2024 data, which can be credited to the comparatively wider date range and therefore less consistent calibration retrieval (leaving us with 444 of 899 anchors). Nonetheless, this shift that we have been noticing around day 35 persists, this time quantifiable as around 30 new anchors reaching 2%, and 20 additional anchors reaching 5% drift thresholds. It appears that a total of around 25 anchors reach the 10% threshold by day 77, and only 1 anchor exceeds 20% over the entire recorded period.

It is encouraging to observe that in the 2024 data, the number exceeding the 20% threshold does not exceed 5 anchors, and in 2026, it is merely 1. Our previous plot appeared failed to visually quantify the number of anchors accumulating the most drift, and to understand the ratio of anchors drifting past a 20% threshold as 4 of 659 anchors is encouraging in grasping the proportion of “erratic” or “volatile” anchors to “stable” anchors that maintain 1-5% drift.->

## 4.2 Anchors IDs that Exceeded 20%, 10%, and 5%

EDIT THIS ITS FROM BEFORE 4 Anchor IDs Exceeded 20%: 7192, 7198, 7207, 7222

21 Anchor IDs Exceeded 10%: 6392, 6398, 6410, 6586, 6718, 6782, 6882, 6887, 6935, 6977, 6994, 7050, 7077, 7078, **7192**, **7198**, **7207**, 7219, **7222**, 7239, 7251

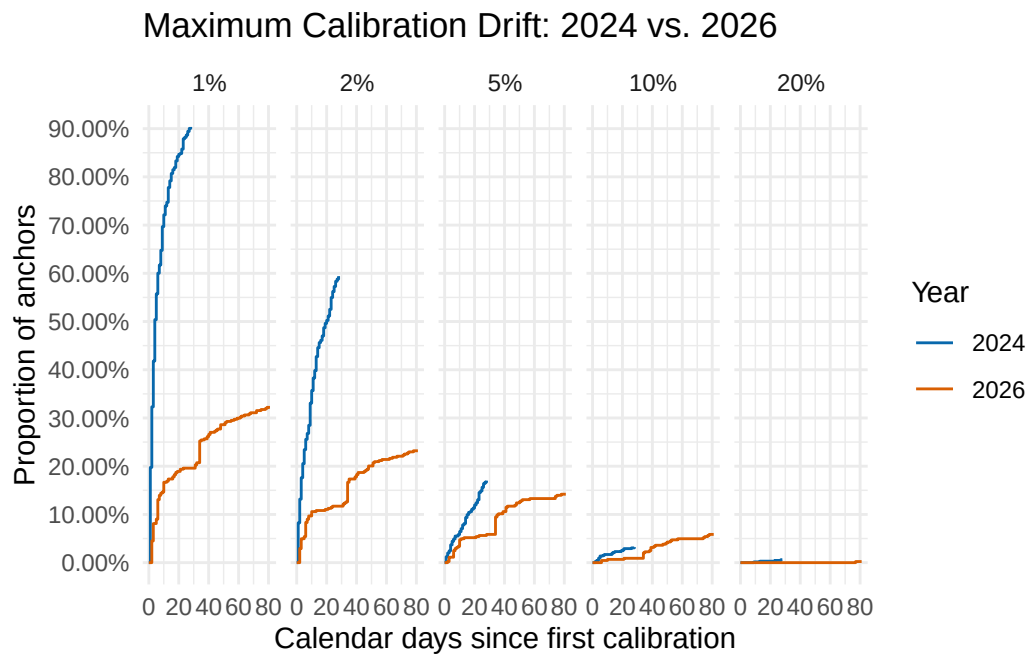
1 Anchor ID Exceeded 20%: 7198

26 Anchor IDs Exceeded 10%: 6313, 6325, 6360, 6509, 6696, 6809, 6841, 6922, 6954, 7003, 7008, 7011, 7097, 7128, 7169, 7181, **7198**, 7216, 7225, 7456, 7502, 7511, 7574, 7616, 7635, 7680

We previously discussed anchor ID 7198 in Table 6, concerning anchors associated with AS3491. The anchor occupied the 90-99.9th percentile range for 27 of 29 days in 2024, and 20 of 80 days in 2026. The other IDs 7192, 7207, and 7222 that exceeded 20% in 2024 and do not appear in the ID summaries from *Section 3.4* and *Section 3.5*. The italicized anchors are all omitted from the *Section 3.4* and *Section 3.5* summaries, and provide new data for us to consider in terms of a new subset of outlier anchors.

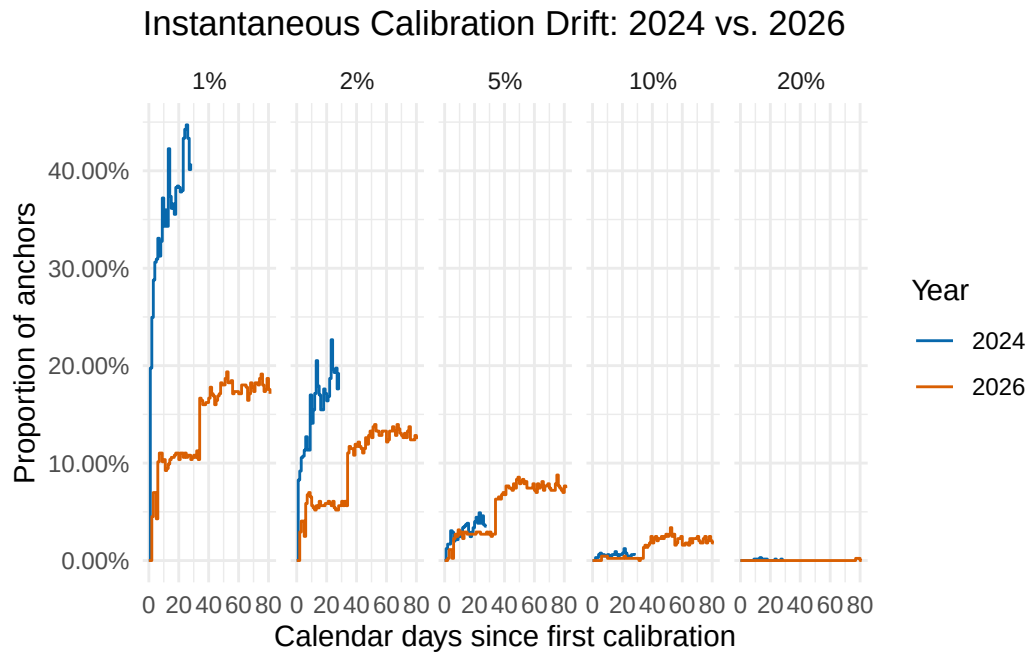
## 5. Direct 2024 vs. 2026 Comparison

### 5.1 Maximum Calibration Drift Threshold



Look at the curves and see if there's a relationship between the two

## 5.2 Instantaneous Calibration Drift Threshold



Compare proportion of instantaneous drift , calculate relationship between curves

### 5.3 Discussion

## 6. Notes for Interpretation

The maximum-drift plots measure whether an anchor has **ever** departed from its initial calibration by a given amount. Consequently, these curves can only increase or remain constant.

The instantaneous-drift plots measure how many anchors are **currently** beyond each threshold. These curves can increase or decrease as anchors move back toward their initial calibration.

A pattern in which maximum drift increases over time while instantaneous drift remains approximately stable suggests that individual anchors experience deviations but that the overall population does not undergo a systematic accumulation of drift.

The 2024 and 2026 comparisons should be interpreted using calendar time. Missing calibration dates in the 2026 dataset remain gaps in the time axis rather than being compressed into consecutive observation numbers.